Computational analysis of single-cell RNA-seq data

Falko Noé
falnoe@ethz.ch
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Plan for today

- Cover the classical scRNA-seq pipeline steps
- Generate and analyse these steps in R/Seurat

- Classical scRNA-seq workflow

Single sample

1. Generation of the count matrix
2. Droplets and doublets detection
3. Quality control on cells and transcripts
4. Cell cycle
5. Feature selection and dimensionality reduction
6. Clustering
7. Cell type annotation
8. *Trajectory analysis*

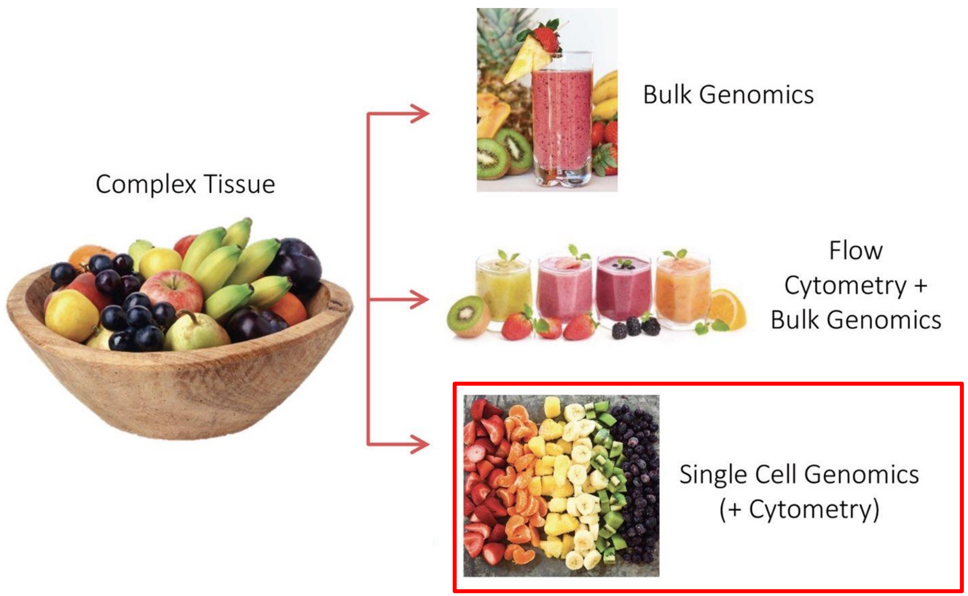
Multiple samples

9. *Integrating and comparing samples*
10. *Differentially expressed genes (DEGs)*
11. *Compositional analyses*

Denoising of the data (steps 1–5)

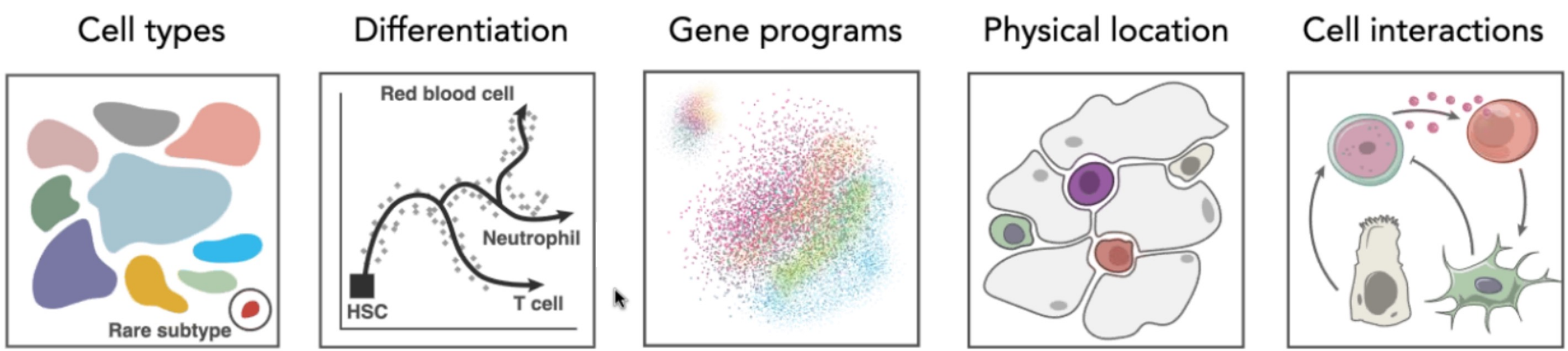
Biological interpretation (steps 6–11)

bulk and single-cell genomics



credit: Aviv regev

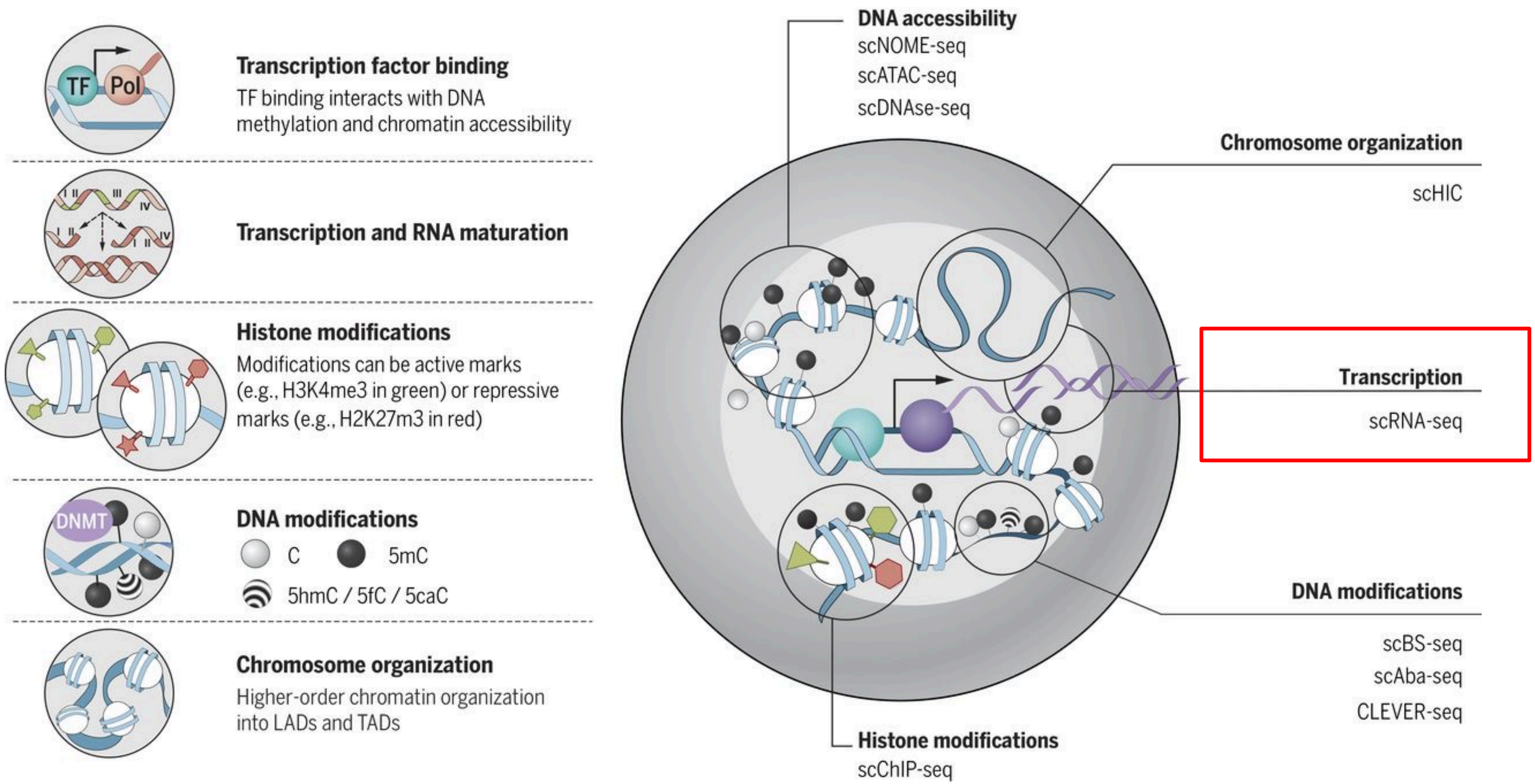
Single-cell genomics addresses fundamental questions



credit: Aviv regev

Single-cell genomics toolkit

Increasing number of single-cell modalities: multi-omics



Challenges of scRNA-seq analysis

- **Large volume of data:** hundreds of thousands of reads for thousands of cells that requires higher amounts of memory to analyze, larger storage requirements, and more time to run the analyses

- **Low depth of sequencing per cell:** For the droplet-based methods of scRNA-seq, the depth of sequencing is shallow, often detecting only 10-50% of the transcriptome per cell. This results in cells showing zero counts for many of the genes (= gene dropouts).

- **Biological variability across cells/samples**

- **Technical variability across cells/samples** (details on next slide)

Technical variability across cells/samples

- **Cell-specific capture efficiency:** Cells will have differing numbers of transcripts captured resulting in differences in sequencing depth
- **Library quality:** Degraded RNA, low viability/dying cells, lots of free-floating RNA, poorly dissociated cells, and inaccurate quantitation of cells
- **Amplification bias**: Not all transcripts are amplified at the same level
- **Batch effects:**
  - Were all RNA isolations performed on the same day?
  - Were all library preparations performed on the same day?
  - Did the same person perform the RNA isolation/library preparation for all samples?
  - Did you use the same reagents for all samples?
  - Did you perform the RNA isolation/library preparation in the same location?

Motivation and challenges of scRNA-seq

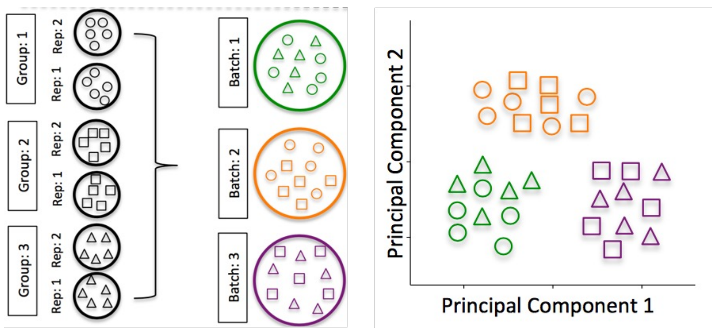
Recommendations

- Do not perform single-cell RNA-seq unless it is **necessary** for the experimental question of interest (bulk RNA-seq could be sufficient, cheaper and more sensitive)

- Understand the **details** of the experimental question you wish to address

- Avoid technical **sources of variability**, if possible. If not:
  - Ensure experimental groups are balanced for demographic and technical factors - avoid having all control samples from one sex/age group or processed in different batches than treatment samples
  - Prepare libraries at same time or **alternate sample groups to avoid batch confounding**
  - The **more replicates the better** when comparing conditions

Example of strong batch effect

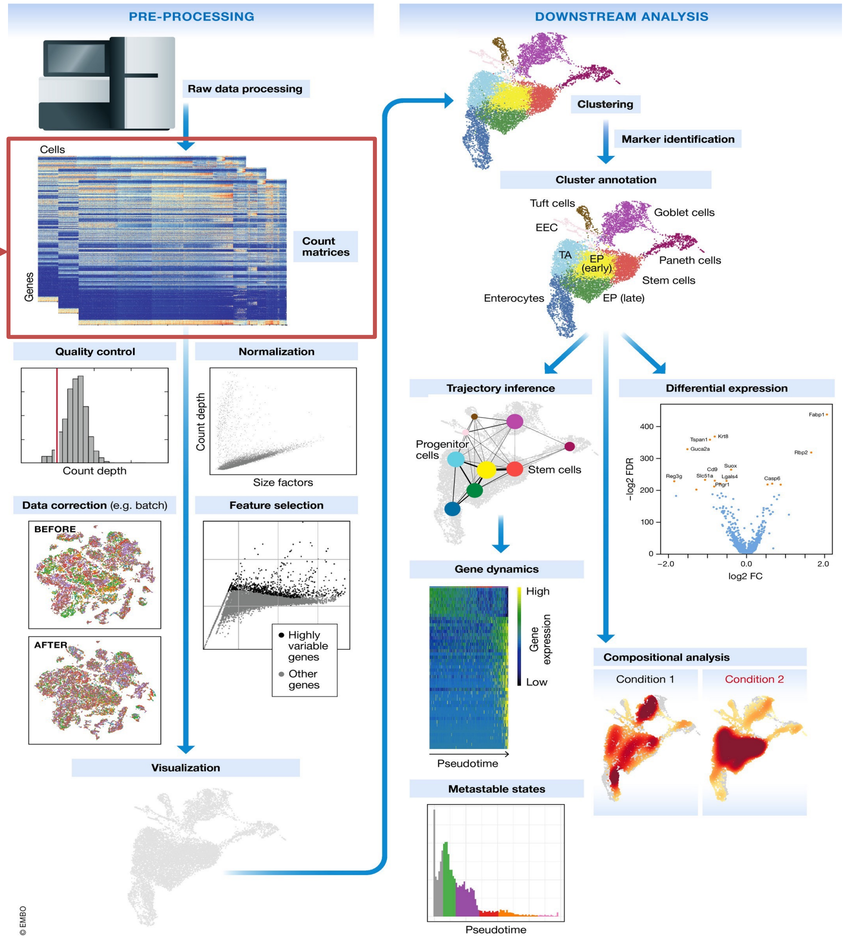


Hicks SC, et al., bioRxiv (2015)

Classical scRNA-seq workflow

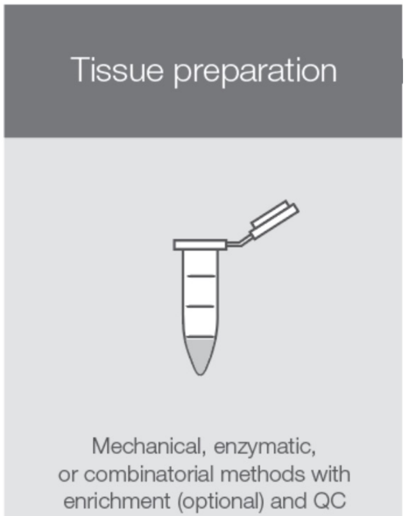
Steps

- Generation of the count matrix
- Empty droplets detection
- Doublet detection
- Quality control: cells and transcripts
- Cell cycle
- Normalization
- Feature selection
- Dimensionality reduction
- Clustering
- Cluster markers
- Cell type annotation
- Trajectory analysis
- Integrating and comparing samples

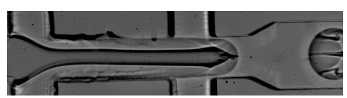


M.D. Luecken, F.J. Theis, "Current best practices in single-cell RNA-seq analysis: a tutorial", Molecular Systems Biology 15(6) (2019): e8746

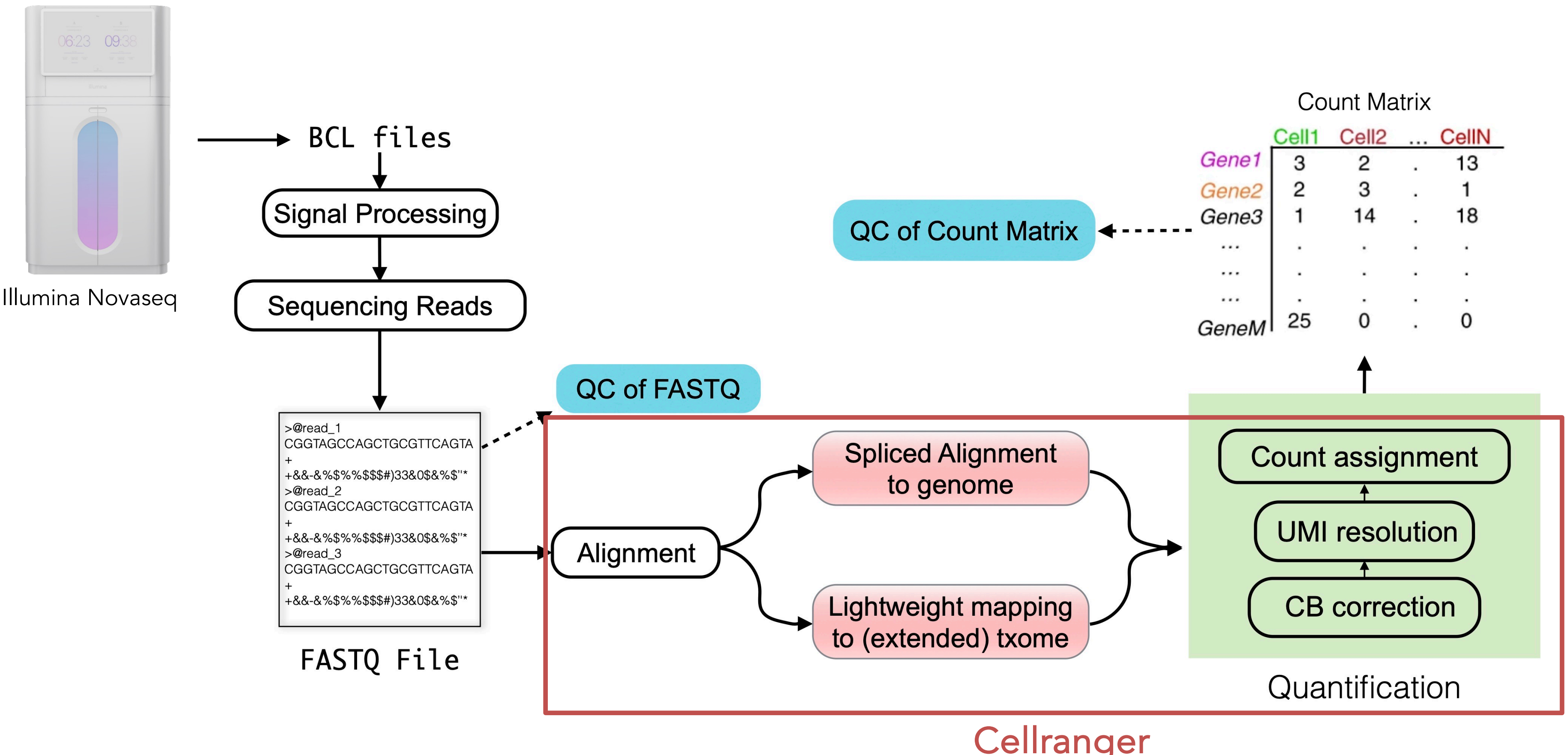
Generation of the count matrix



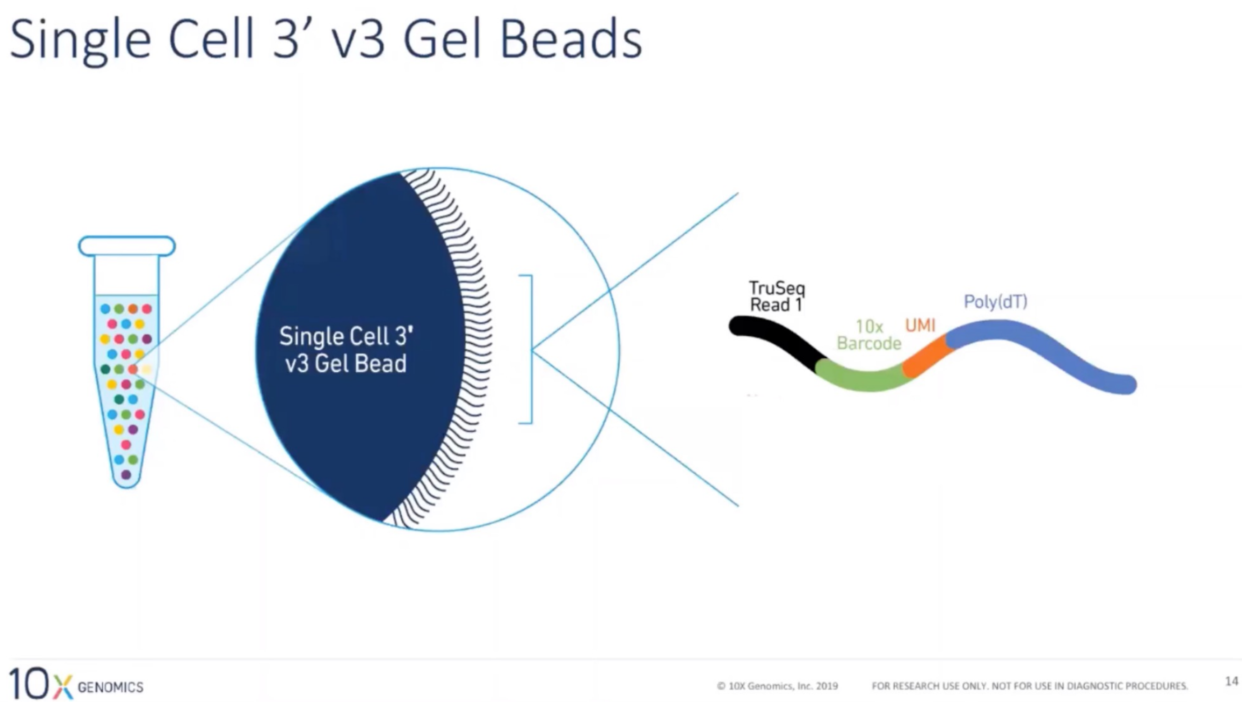
source: illumina



Generation of the count matrix



10x Genomics read structure

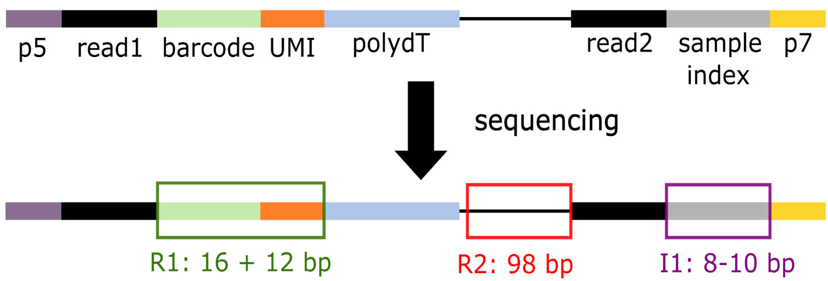


After second strand synthesis

Generation of the count matrix

The following are required for proper quantification at the cellular level (Example 10x Genomics):

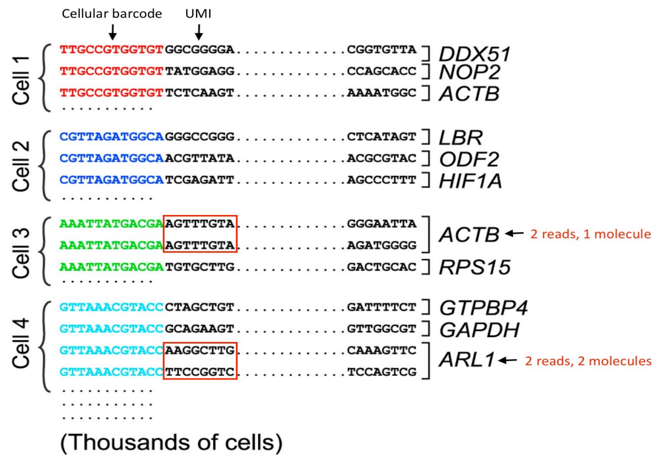
- Sample Index
- Cellular barcode (CB)
- Unique molecular identifier (UMI)
  - 1 UMI = 1 RNA molecule
- Sequencing Read1: 16 bp 10X Cellular Barcode + 12 bp UMI
- Sequencing Read2: cDNA fragment



10x Genomics sequence read structure

Only capture 98bp of 3' of the genes!

Use of UMIs for droplet-based methods

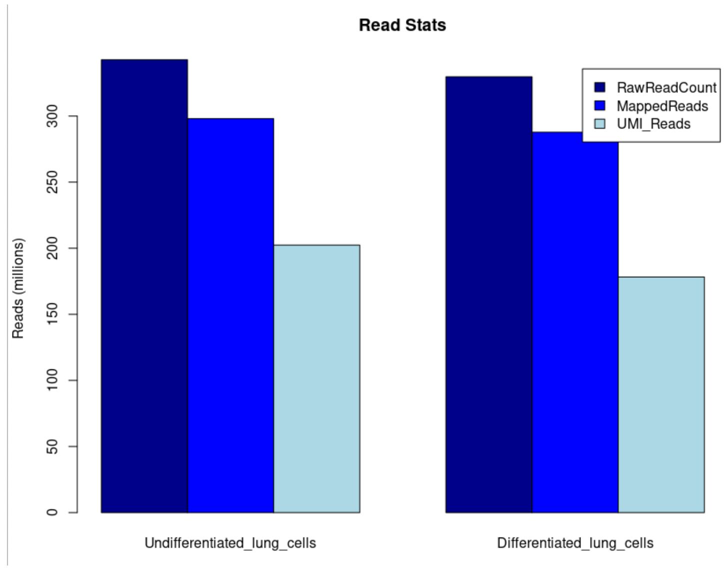


Generation of the count matrix

1. Demultiplexing the reads using the cell barcodes
2. Mapping/pseudo-mapping to transcriptome
3. Collapsing UMIs (=Deduplication) and quantification of genes

→ All steps performed by CellRangerApp via SUSHI

`Why count UMI (and not read alignments?)`
Unique Molecular Identifier:
- Identifies each molecule (i.e. sequence) uniquely
- Molecules from a common PCR template → carry the same UMI
- By counting UMI: correct for PCR duplicates



Quiz

- Q1: What is a UMI and why is it important for scRNA-seq quantification?
- Q2: What information is contained in Read1 vs Read2 in 10x Genomics sequencing?

Quiz

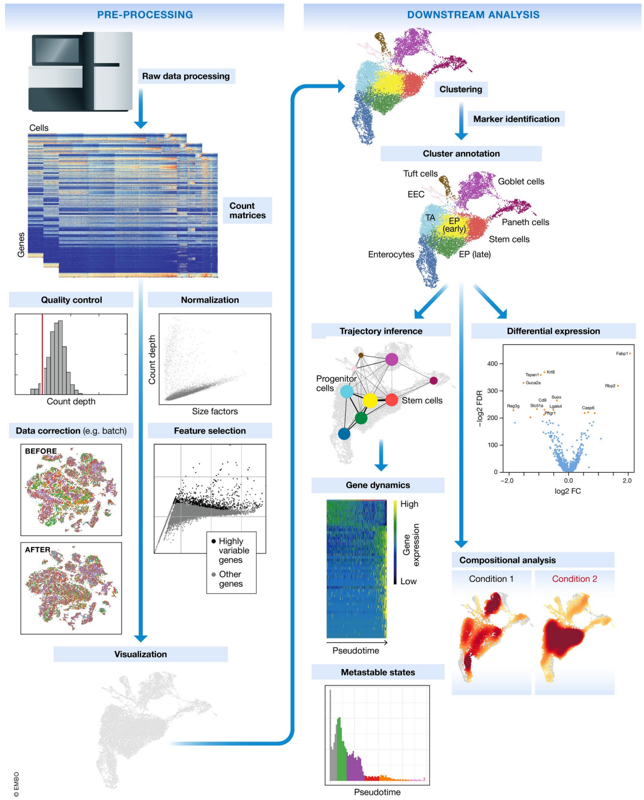
- Q1: What is a UMI and why is it important for scRNA-seq quantification?
    - Unique Molecular Identifier (UMI) is a short random sequence that tags each original RNA molecule. It's important because it helps distinguish between original molecules and PCR duplicates, enabling more accurate quantification.
- Q2: What information is contained in Read1 vs Read2 in 10x Genomics sequencing?
    - Read1: Contains the 16bp cellular barcode + 12bp UMI (identifies which cell and which original molecule)
    - Read2: Contains the actual cDNA sequence fragment (identifies which gene was expressed)

Single-cell RNA workflow

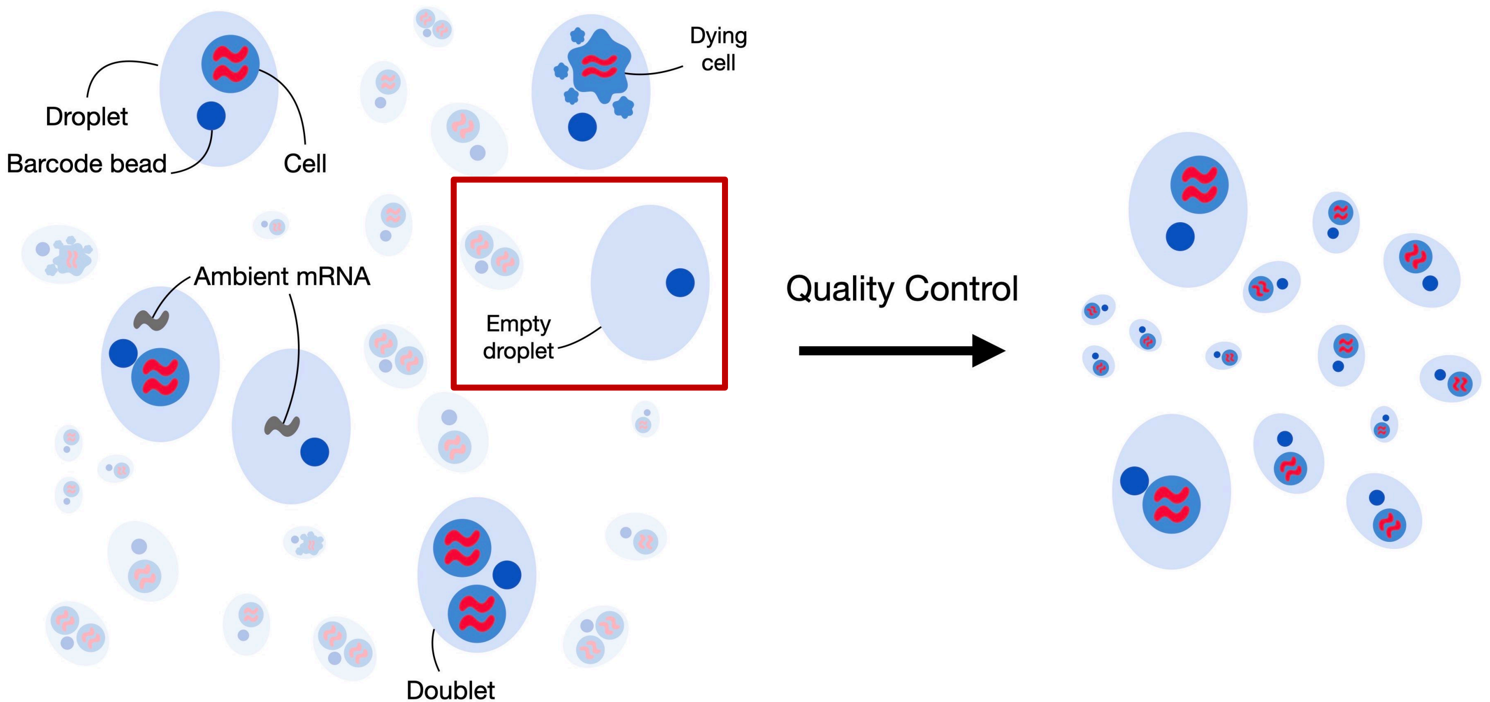
Steps

- Generation of the count matrix
- Empty droplets detection

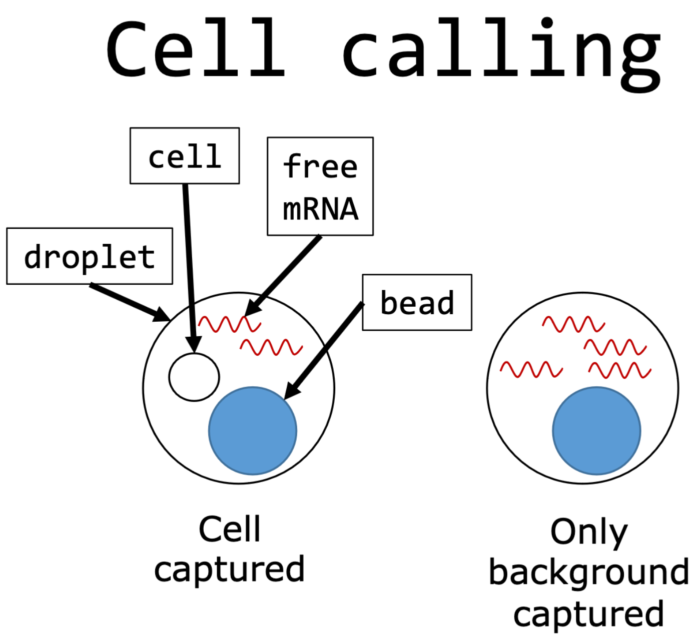
Cellranger does both



Quality control



Empty droplet detection via CellRanger



Empty droplets detection / cell calling

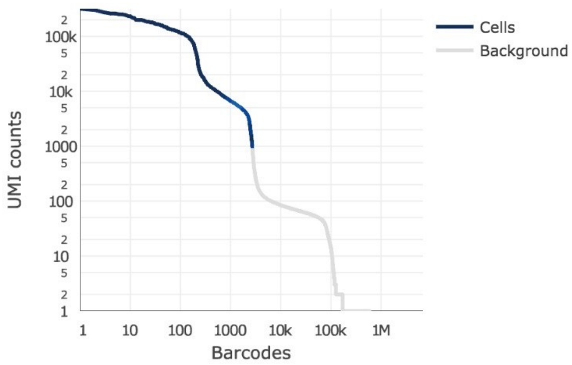
Droplet-based data: we have no prior knowledge about whether a particular library (i.e., cell barcode) corresponds to cell-containing or empty droplets. Thus, we need to call cells from empty droplets based on the observed expression profiles.

Challenge: Empty droplets can contain ambient (i.e., extracellular) RNA that can be captured and sequenced, resulting in non- zero counts for libraries that do not contain any cell. Low RNA content cells could be discarded.

Solution:

- **Test whether the expression of each barcode is significantly different from the ambient RNA pool. Any significant deviation indicates that the barcode corresponds to a cell containing droplet.**

→ Performed within CellRangerApp



Quiz

- Q1: How can you identify if a droplet contains ambient RNA versus cell RNA
- Q2: What is the "knee plot" in empty droplet detection and what does it show?

Quiz

- Q1: How can you identify if a droplet contains ambient RNA versus cell RNA
    - Droplets with only ambient RNA typically show lower UMI counts and different gene expression patterns compared to cell-containing droplets.

- Q2: What is the "knee plot" in empty droplet detection and what does it show?
    - A log-log plot of UMI counts versus barcode rank that reveals a characteristic "knee" shape where cell-containing droplets (high UMI, steep curve) transition to empty droplets (low UMI, gradual decline).

Cellranger output – web summary

Cell Ranger • multi

Cells

Library

Experimental Design

More information on web summary interpretation

10x GENOMICS

Sample ID

control-spleen_Plate_3455_VDJ_GEX-cellRanger

Pipeline Version

cellranger-8.0.0

Gene Expression VDJ-T

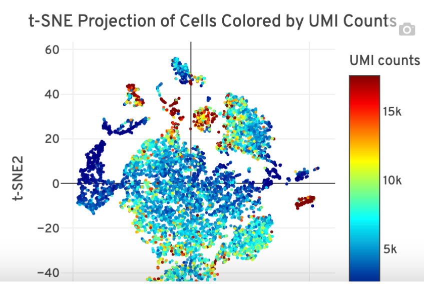
Chemistry	Single Cell 5' R2-only
Reference Path	/srv/GT/reference/Mus_musculus/GENCODE/GRCm39/Annotation/Release_M31-2023-01-30/Genes/genes_10XGEX_SC_Mt_rRNA-Mt_tRNA-protein_coding-rRNA-tRNA_Index
Transcriptome	genes_10XGEX_SC_Mt_rRNA-Mt_tRNA-protein_coding-rRNA-tRNA_Index-
Include Introns	true

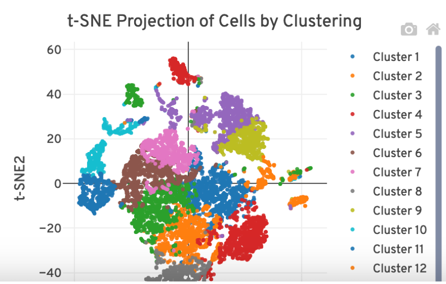
Cells

Cells	Mean reads per cell	Median genes per cell	Total genes detected	Median UMI counts per cell	Confidently mapped reads in cells
10,345	45,216	2,128	18,881	5,550	92.76%

t-SNE Projection

Clustering Type: Graph-based





Cellranger output – web summary

Cell Statistics

Physical library ID	Estimated number of cells	Mean reads per cell
GEX_1	10,345	59,454

Sequencing Metrics

Fastq ID	Number of reads	Number of short reads skipped	Q30 barcodes	Q30 UMI	Q30 RNA read
control-spleen_Plate_3455_VDJ_GEX	615,054,882	0	96.2%	97.0%	92.8%

Mapping Metrics (Amongst All Reads in Library)

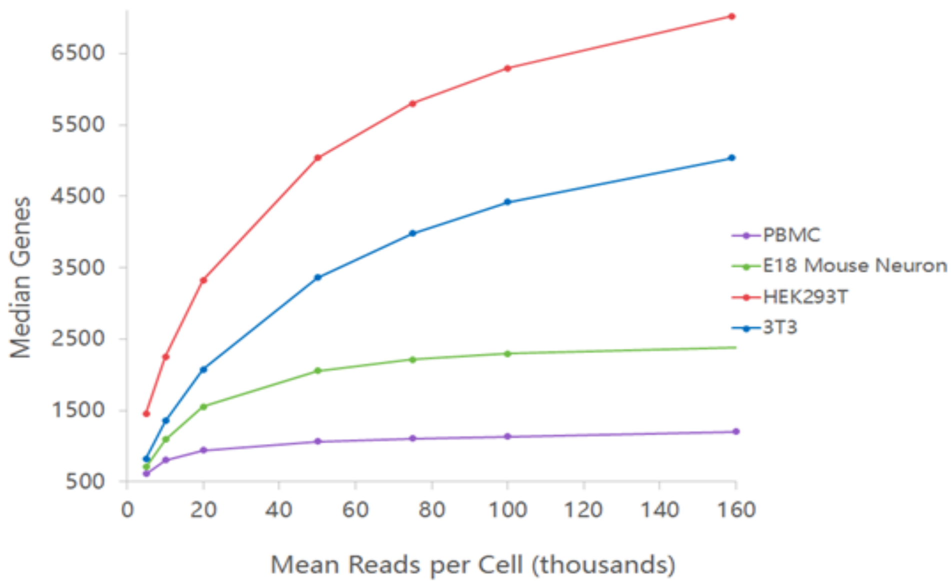
Physical library ID	Number of reads in the library	Mapped to genome	Confidently mapped to genome	Confidently mapped to transcriptome	Confidently mapped to intronic regions	Confidently mapped to exonic regions	Confidently mapped to intergenic regions	Confidently mapped antisense
GEX_1	615,054,882	87.55%	82.94%	63.86%	10.65%	64.16%	8.13%	10.38%

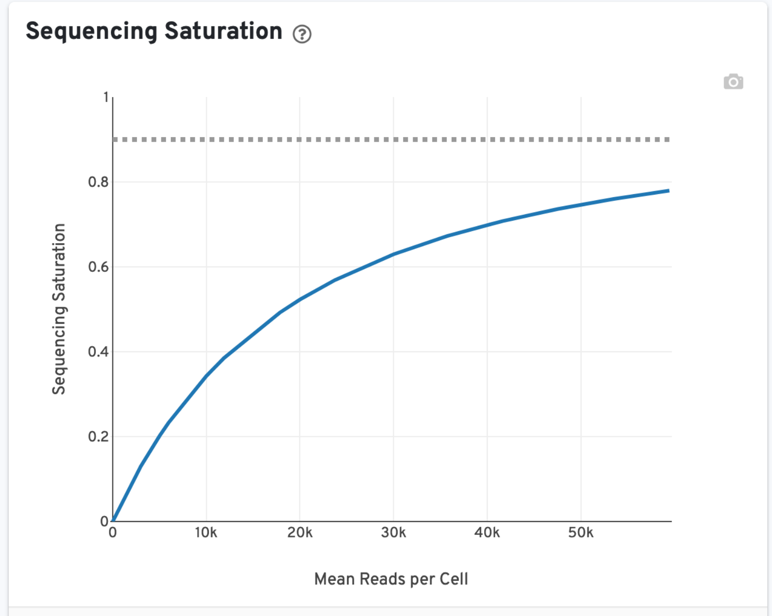
Metrics Per Physical Library

Physical library ID	Number of reads	Valid barcodes	Valid UMIs	Sequencing saturation	Confidently mapped reads in cells	Mean reads per cell
GEX_1	615,054,882	86.88%	99.64%	77.98%	92.76%	59,454

Cellranger output – sequencing saturation

- **Higher** saturation (>80%) means you've captured most available transcripts
- **Lower** saturation suggests deeper sequencing could yield more unique molecules.





Sequencing Saturation =

$$1 - \frac{\#deduplicatedReads}{\#totalReads}$$

Hands-On analysis with Seurat in Renku

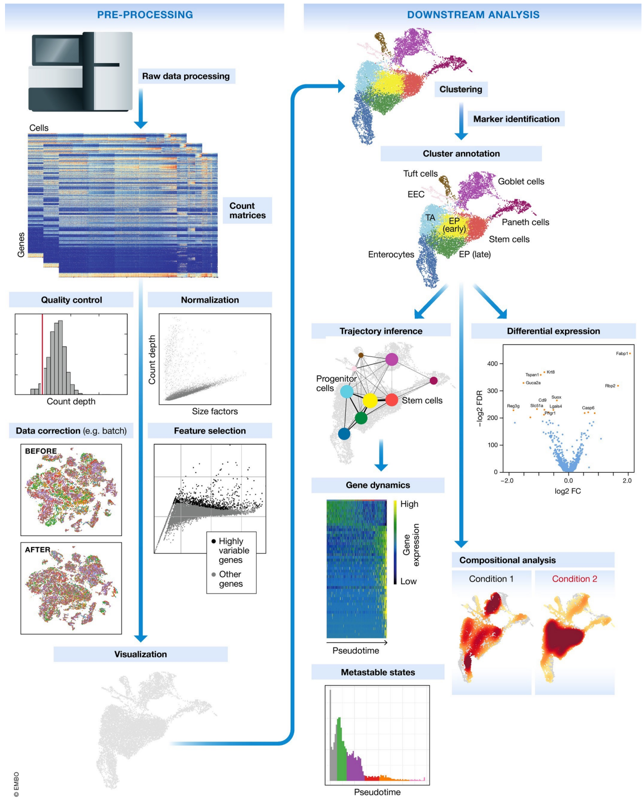
CellRanger

Single-cell RNA workflow

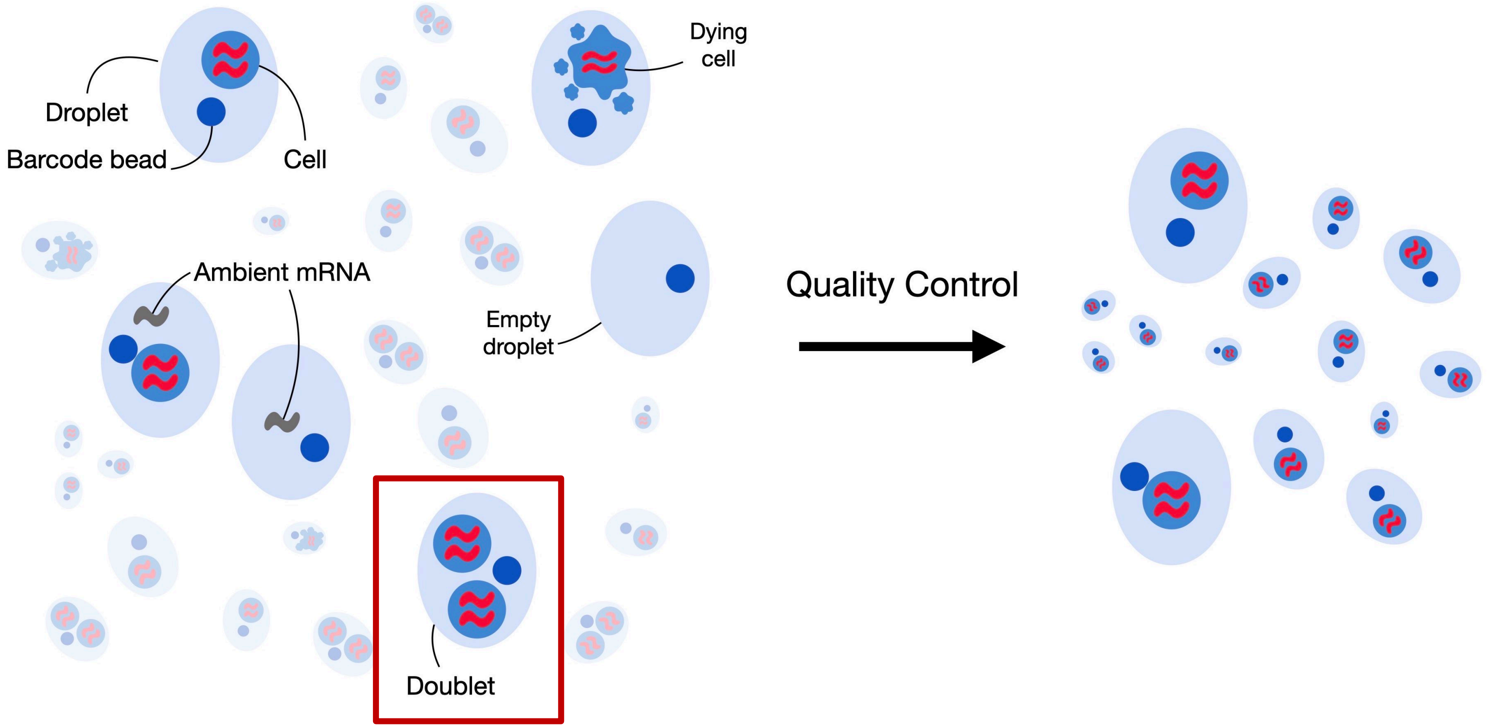
Steps

- Generation of the count matrix
- Empty droplets detection
- Doublet detection

Cellranger



Quality control

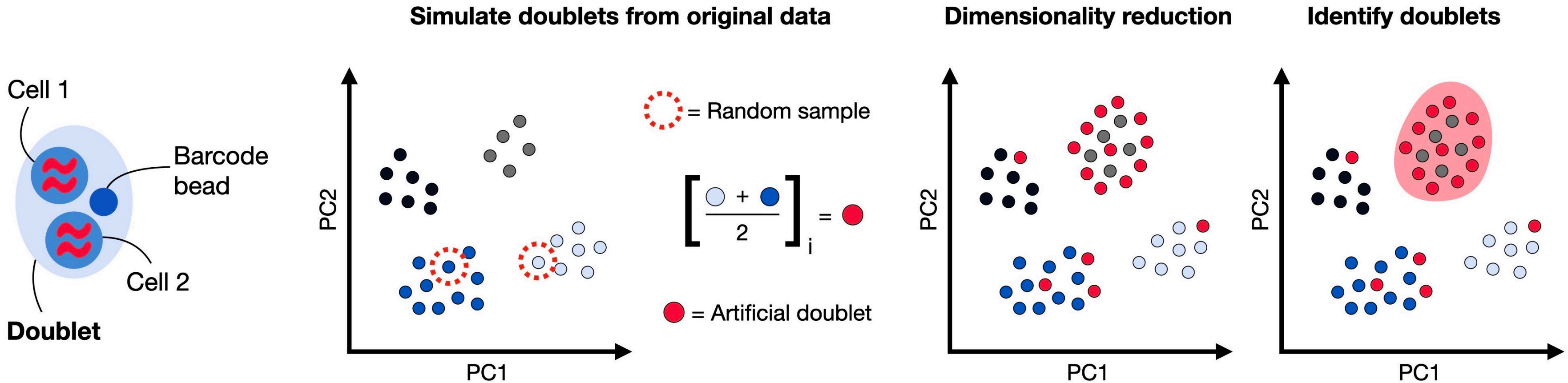


Doublet detection

- Doublets/Multiplets arise when two or more cells are captured within the same reaction → **hybrid transcriptome (present in microfluidics systems)**

- Doublets can have varying consequences for downstream analyses
  - Spurious clusters
  - Spurious differentiation paths between cell types

How can we identify doublets?



→ R-Bioconductor package 'ScDblFinder'

Doublet detection and removal

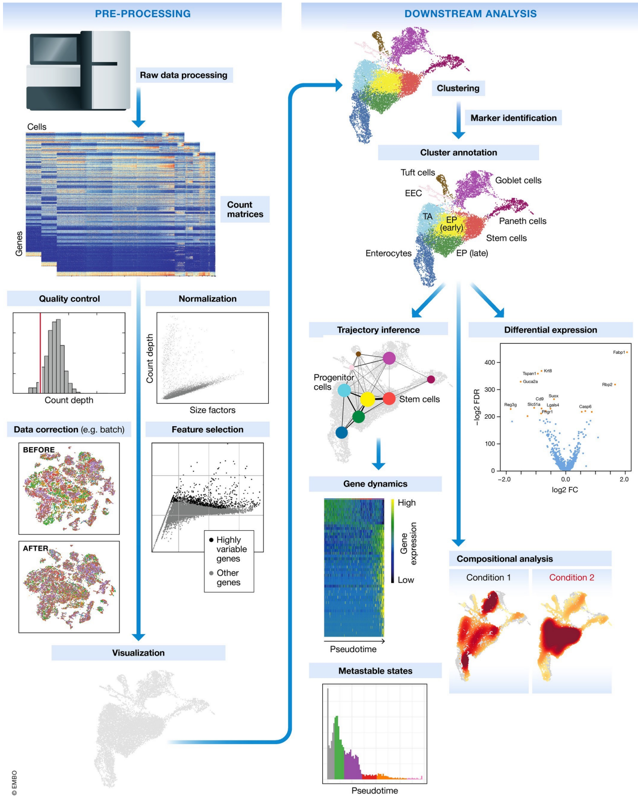
Further comments and suggestions

- Always remove doublets => Doublets are fairly frequent in scRNAseq datasets, with estimates ranging from 1 to 10% depending on the platform and cell concentration used
- All cells from a cluster with a large average doublet score should be considered suspect, and close neighbours of problematic clusters should also be treated with caution
- In contrast, a cluster containing a small proportion of high-scoring cells is probably safe provided that any interesting results are not being driven by those cells.
- **Only within the same batch**: Doublet detection procedures should only be applied to libraries generated in the same experimental batch.
- **Cell loading matters: Larger number of loaded cells will drive to more doublets. (around 1% per 2k cells loaded)**

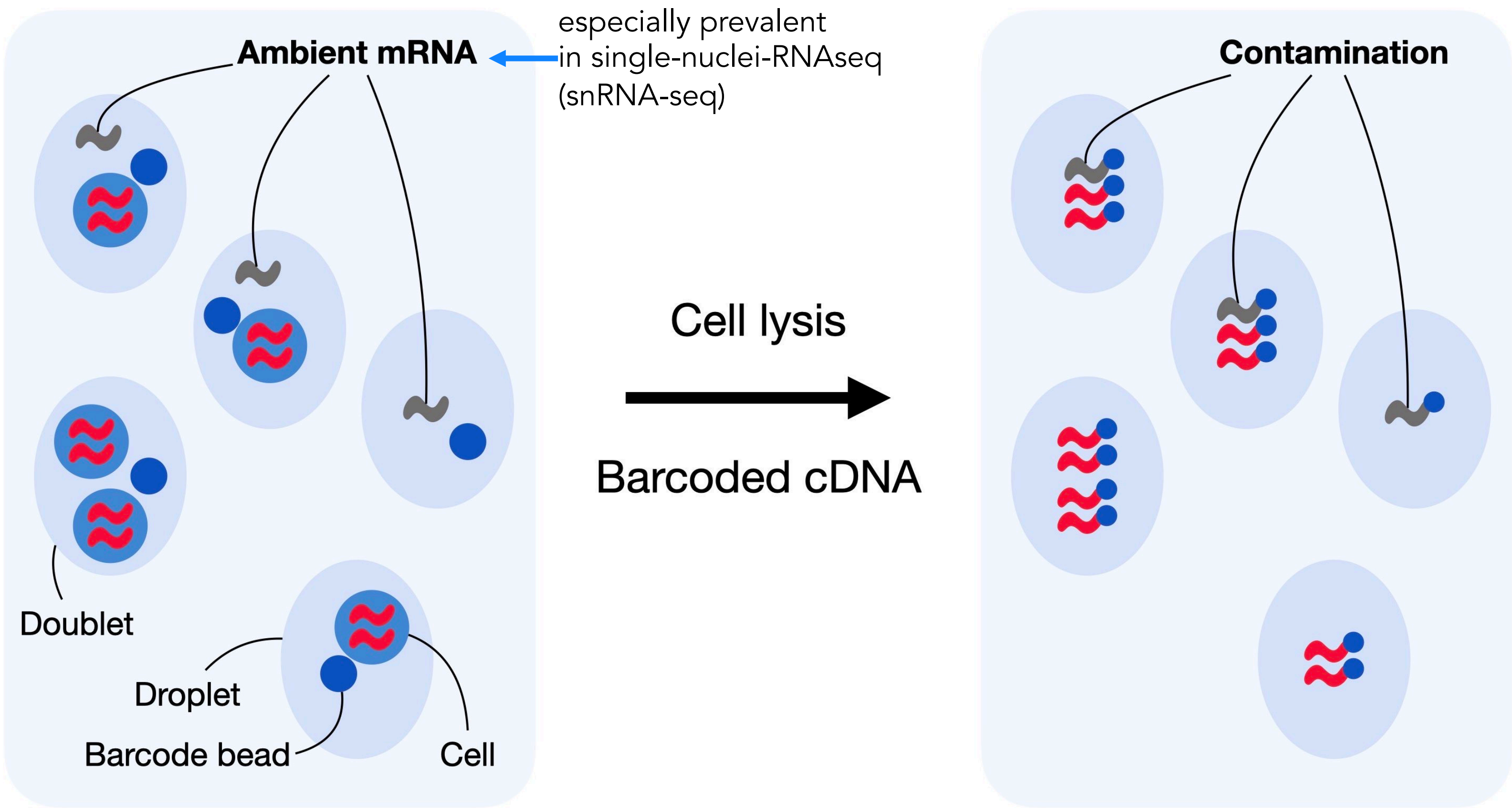
Single-cell RNA workflow

Steps

- Generation of the count matrix
- Empty droplets detection
- Doublet detection
- Ambient RNA removal



Ambient RNA contamination

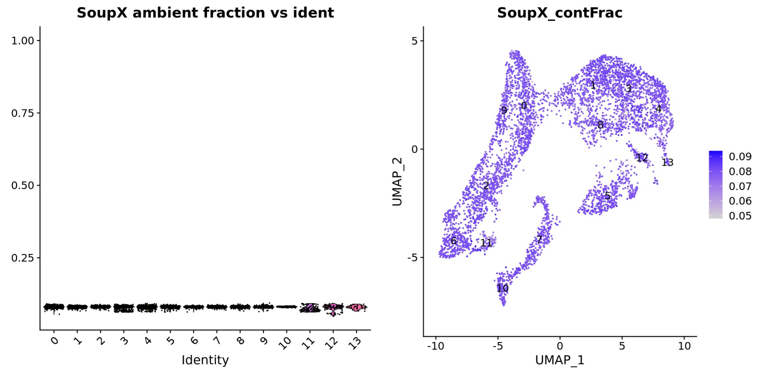


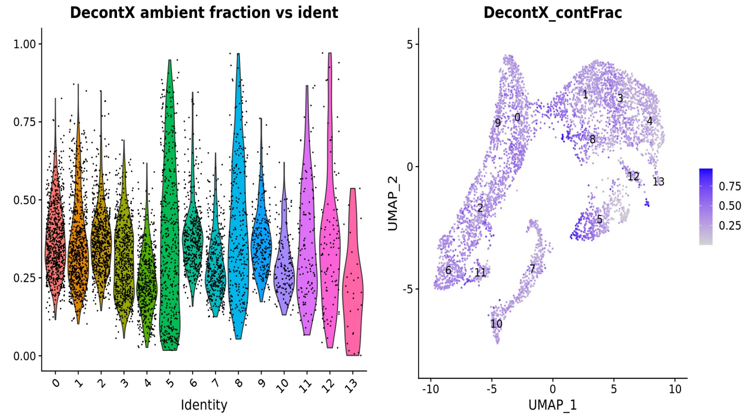
Ambient RNA contamination

- All droplet based single cell RNA-seq experiments also capture ambient mRNAs present in the input solution along with cell specific mRNAs of interest
- Result of **dissociation** procedures or other library preparation processes
- Can vary hugely between experiments (2% - 50%)
- Solid tumours, low-viability and **single-nuclei** suspensions tend to produce higher contamination fractions

→ 3 tools are part of our pipeline: SoupX,DecontX and Cellbender

→ Most recent tool **Cellbender** works better

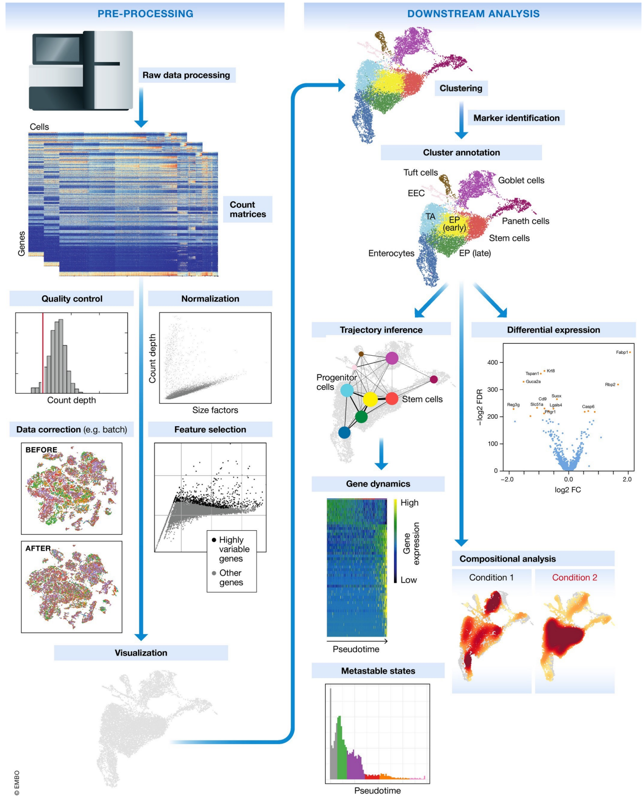




Single-cell RNA workflow

Steps

- Generation of the count matrix
- Empty droplets detection
- Doublet detection
- Ambient RNA removal
- Quality control: cells and transcripts

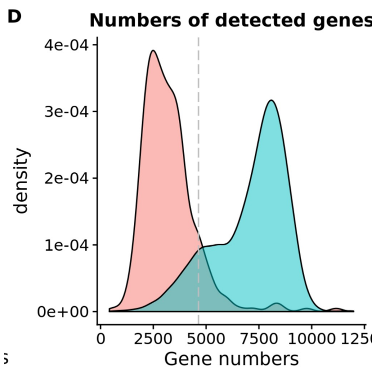


Quality control

Low quality cells can arise from a variety of sources such as **cell damage** during dissociation and library preparation. They can contribute to misleading results in downstream analysis. We need to remove these cells at the beginning of the analysis.

Choice of QC metrics

- library size (#UMIs per cell)
- mitochondrial read fraction per cell

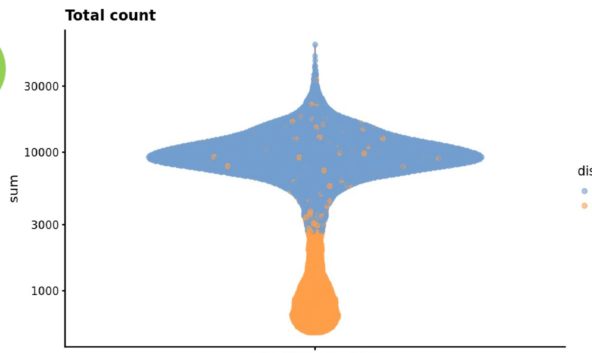


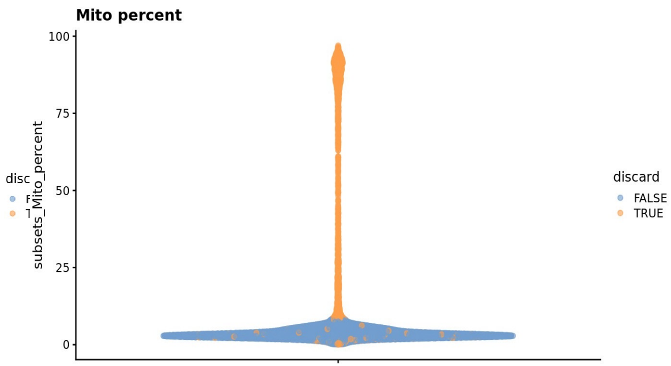
Identification low-quality cells

- **fixed** thresholds → apply fixed thresholds on the QC metrics
- **adaptive** thresholds → exclude cells that differ considerably from most other cells based on the QC metrics based on the median absolute deviation (**MAD**) from the median value of each metric across all cells

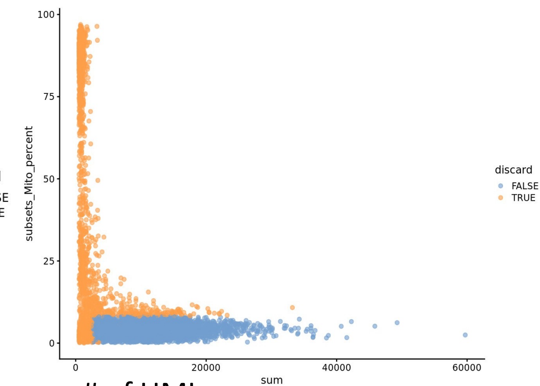
Quality control

Diagnostic plots: Did the filtering strategy work?

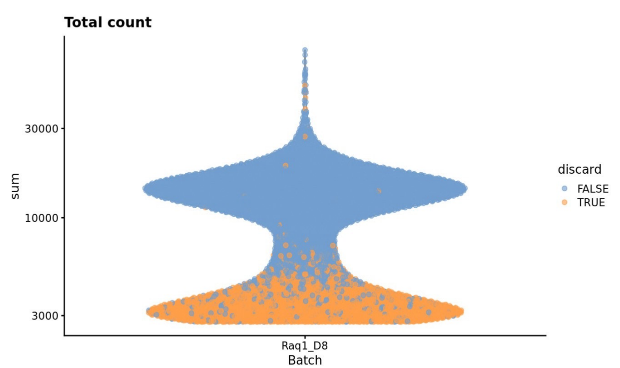


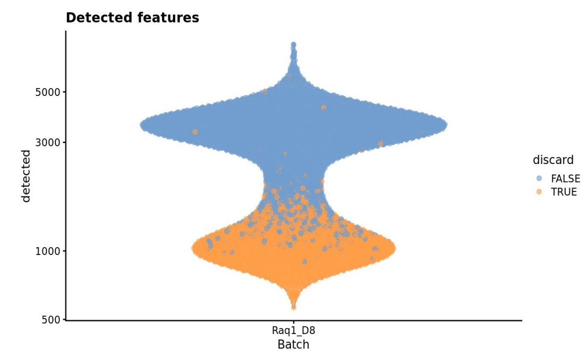


%mito genes



# of UMI





Quality control

Suggestions

- Have a good idea of your expectations for the **cell types** to be present prior to performing QC
- Always perform filtering **before** downstream analysis
- Use **adaptive** thresholds and check for normal distributions using diagnostic plots
- In case of getting large proportion of cells in another mode, use more **relaxed** thresholds for automatic outlier detection or use fixed thresholds

Quality control on transcripts/genes

- Filter out genes that are not expressed in more than a few cells and are thus not informative

Quiz

- Q1: Why do we look at mitochondrial gene percentage during QC?
- Q2: What are adaptive thresholds in QC and when should they be used?

Quiz

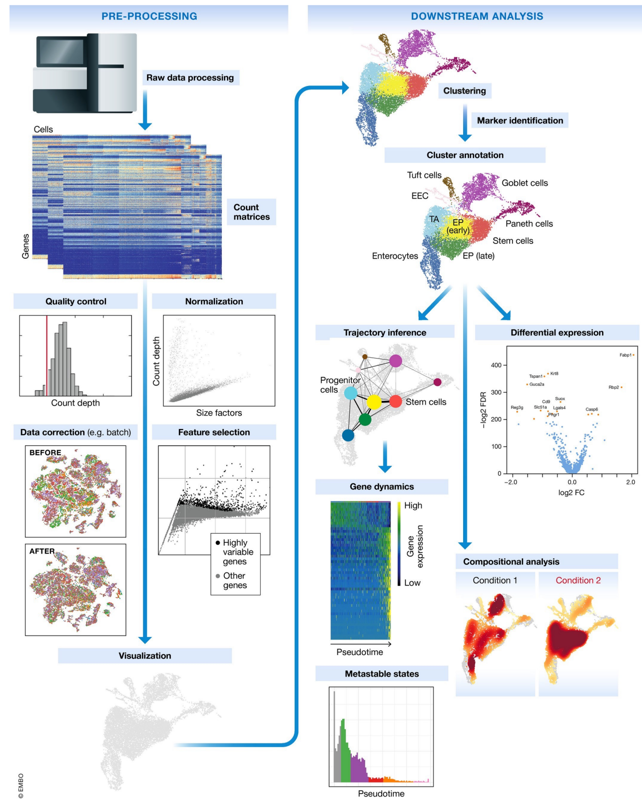
- Q1: Why do we look at mitochondrial gene percentage during QC?
  - High mitochondrial gene percentage often indicates cell stress or death, as dying cells tend to lose cytoplasmic RNA while maintaining mitochondrial RNA.

- Q2: What are adaptive thresholds in QC and when should they be used?
  - Adaptive thresholds are based on the distribution of QC metrics across all cells (using MAD from median), rather than fixed cutoffs. They should be used when the data shows normal distributions and you want to account for dataset-specific characteristics.

Single-cell RNA workflow

Steps

- Generation of the count matrix
- Empty droplets detection
- Doublet detection
- Ambient RNA removal
- Quality control: cells and transcripts
- Cell cycle



Cell cycle

Motivation

- **Remove uninteresting variation** due to differences in cell cycle state
- Assess whether there are differences in proliferation between subpopulations or across treatment conditions

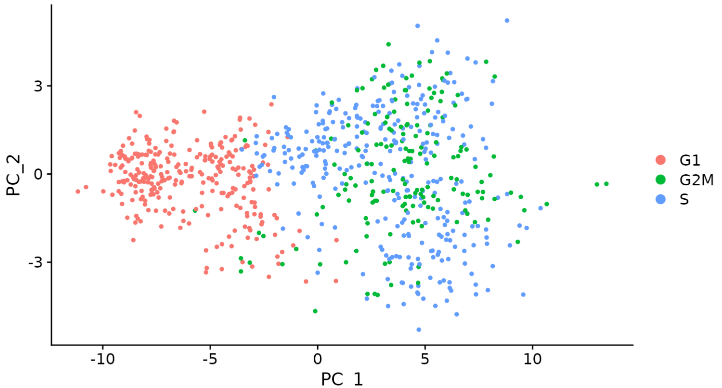
Cell cycle assignment

- Specialized case of cell annotation
- Use of reference datasets containing cells with known cycle phases to identify phase-specific markers from genes with annotated roles in cell cycle to annotate our own dataset

Suggestions

- Don't forget about cell cycle, as some clusters might be driven by cell-cycle
- Don't regress out the cell-cycle signal, but rather work with it

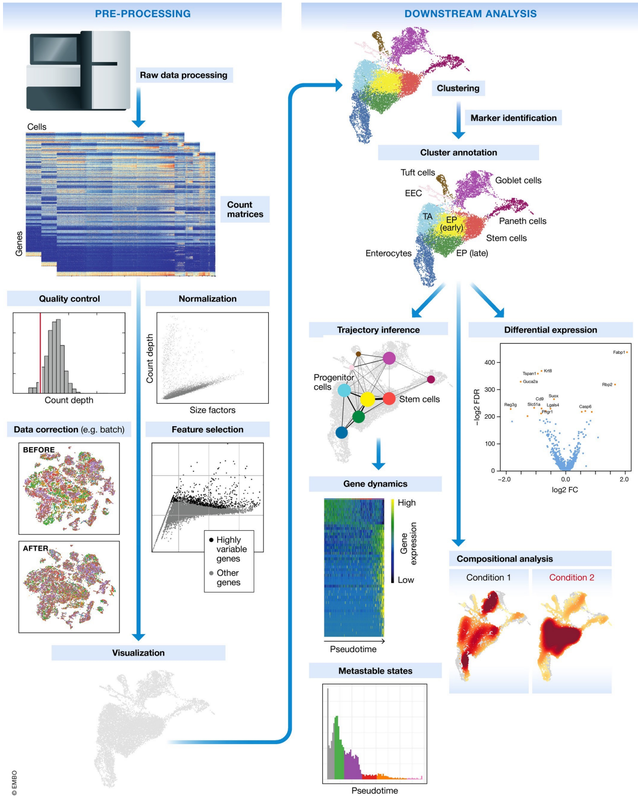
→ Cell cycle phase prediction with R package 'Scran' in our pipeline



Single-cell RNA workflow

Steps

- Generation of the count matrix
- Empty droplets detection
- Doublet detection
- Ambient RNA removal
- Quality control: cells and transcripts
- Cell cycle
- Normalization



Normalization

Motivation

Systematic differences in sequencing coverage between libraries occur because of:

- low input material
- differences in cDNA capture
- differences in PCR amplification

Normalisation **removes** such differences so that differences between cells are less technical and more biological, allowing meaningful comparison of expression profiles between cells.

Normalisation and **batch-correction** have different aims:

Normalisation addresses technical differences only,
while batch correction considers both technical and biological differences.

Normalization – log normalization (Seurat default)

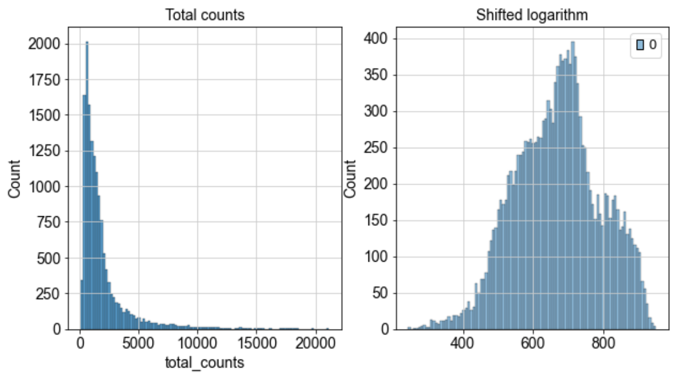
Seurat Normalization Step by Step

1. Raw Counts

	Gene1	Gene2	Total
Cell1	10	5	**15**
Cell2	20	10	**30**
Cell3	5	2	**7**

Values are more comparable across cells after normalization

Log normalization transformation reduces the range of values



Before normalization After normalization

Quiz

- Q1: What is the difference between normalization and batch-correction?
- Q2: Why might traditional RNA-seq normalization methods not work well for scRNA-seq?

Quiz

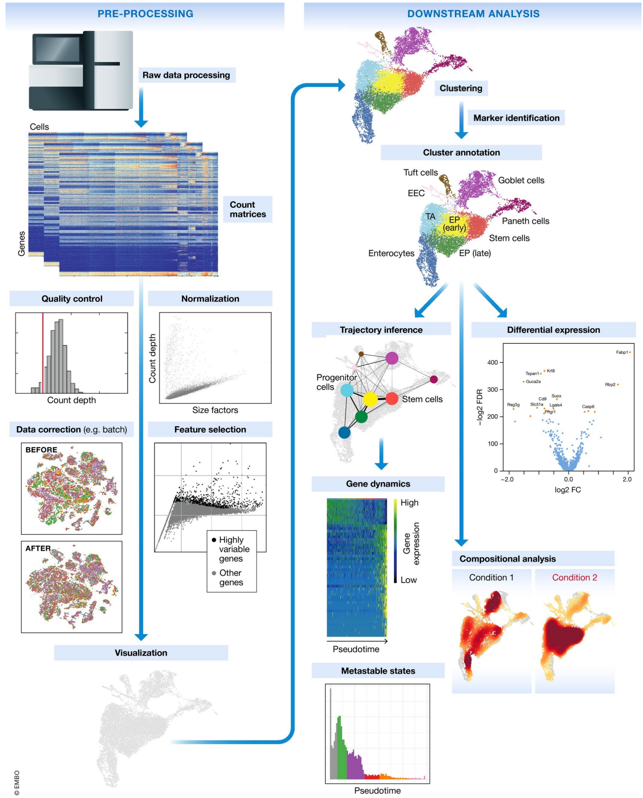
- Q1: What is the difference between normalization and batch correction?
  - Normalization = cell-to-cell technical differences (within the same experiment)
  - Batch correction = experiment-to-experiment systematic differences (between different batches/studies)

- Q2: Why might traditional RNA-seq normalization methods not work well for scRNA-seq?
  - Non-full-length 5′ and 3′ methods do not have a length-bias (only ~90bp of the ends are captured). Traditional RNAseq methods such as TPM and RPKM take gene length into account while, so we risk overcorrecting the counts.

Single-cell RNA workflow

Steps

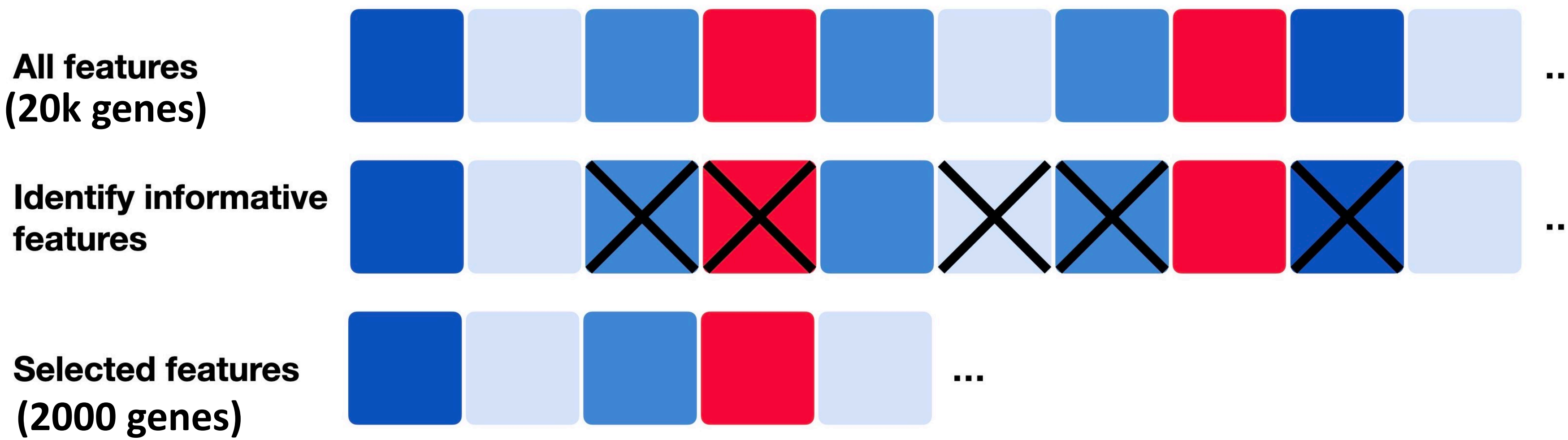
- Generation of the count matrix
- Empty droplets detection
- Doublet detection
- Ambient RNA removal
- Quality control: cells and transcripts
- Cell cycle
- Normalization
- Feature selection



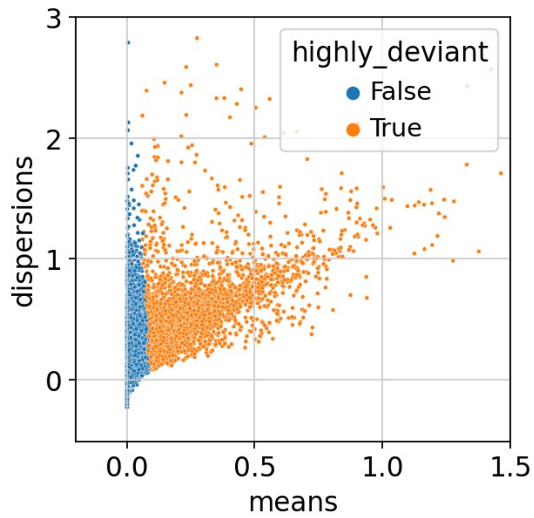
Feature selection (= gene selection)

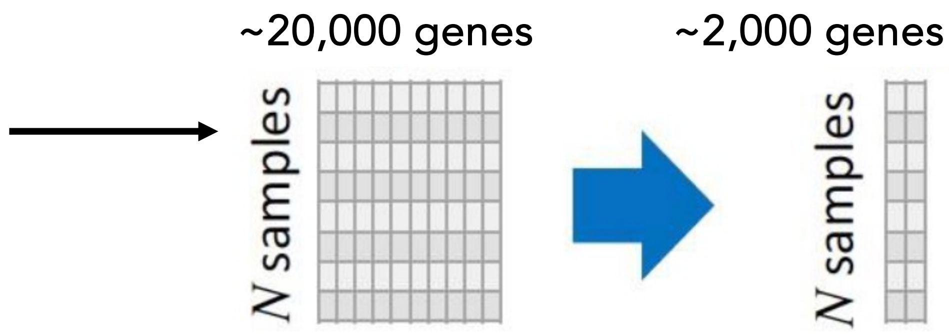
Motivation

- Ease the computational burden on downstream analysis tools
- Reduce the noise in the data



Feature selection





We go from ~20,000 genes to ~2,000 HVGs (Highly Variable Genes)

→ how to represent this?

→ can we reduce this number even more?

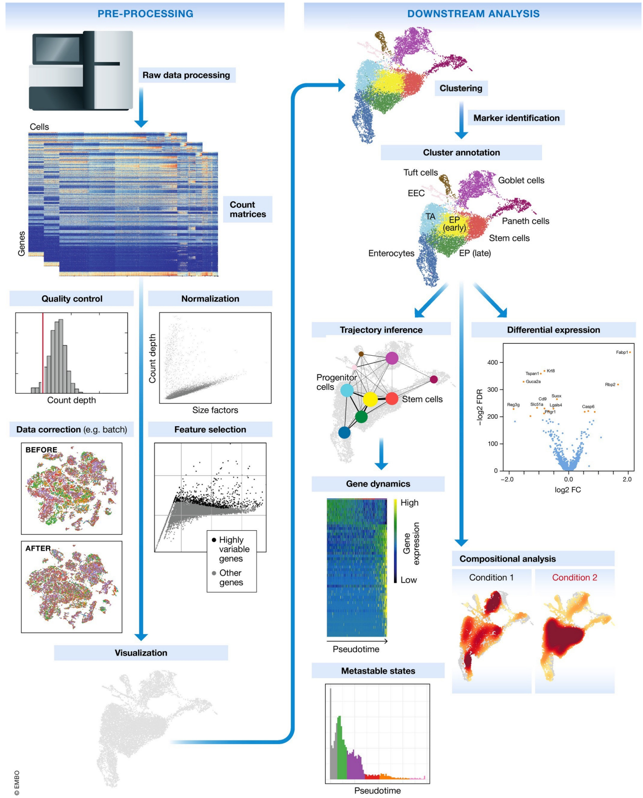
Hands-On analysis with Seurat in Renku

QC

Single-cell RNA workflow

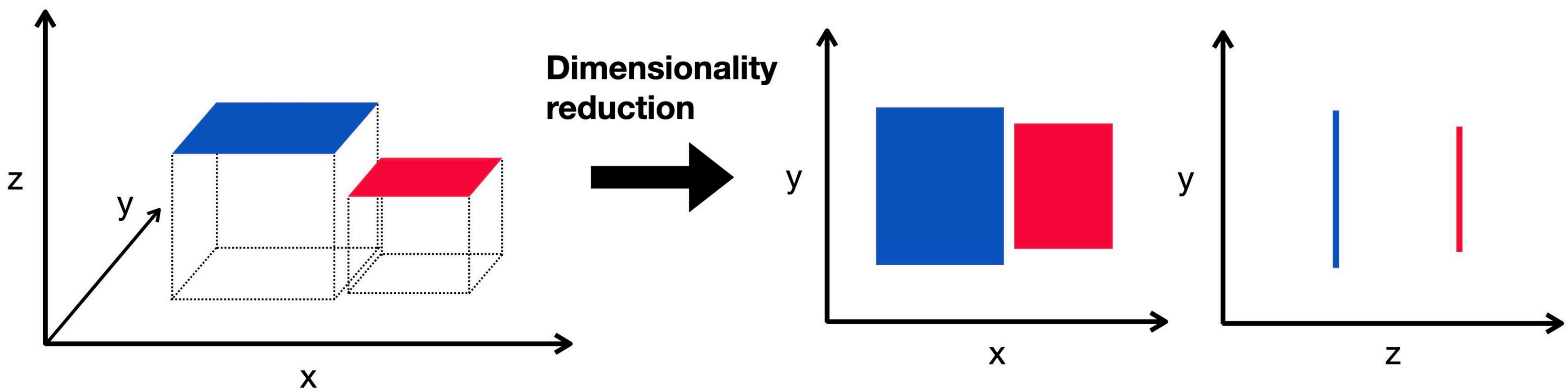
Steps

- Generation of the count matrix
- Empty droplets detection
- Doublet detection
- Ambient RNA removal
- Quality control: cells and transcripts
- Cell cycle
- Normalization
- Feature selection
- Dimensionality reduction



Dimensionality reduction - visualization

How to represent high dimensional data in 2d (=visible to human eyes)?

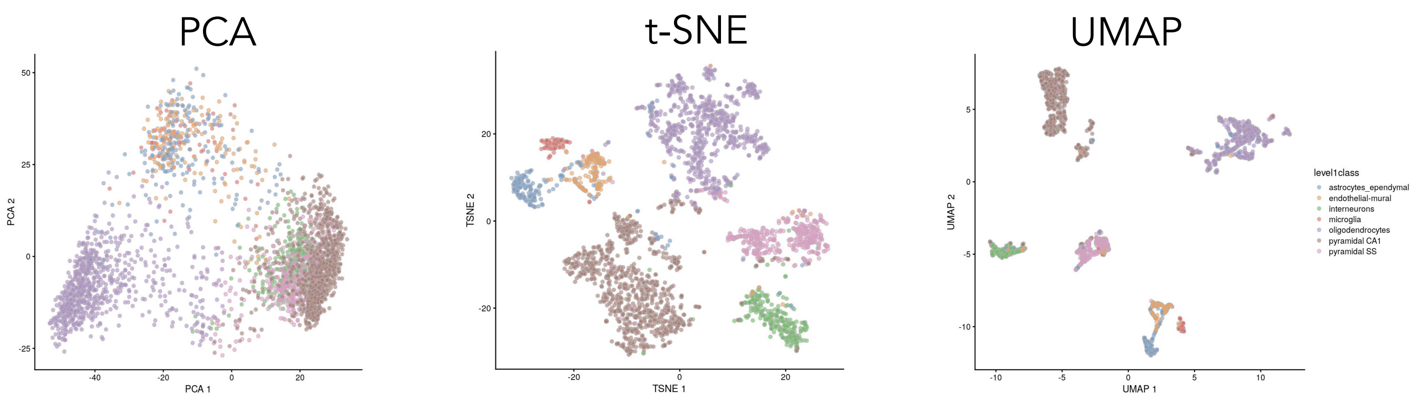


Original space is ~20,000 dimensions

(~20,000 genes)

Dimensionality reduction - visualization

1. **PCA**: is only able to capture the variation along line by each PC. It can't efficiently pack d dimensions into the first 2 PCs.
2. **t-SNE** (T-Stoachastic Neighbour Embedding): is not restricted to linear transformations and it is not forced to represent distances between distinct populations.
3. **UMAP** (Uniform Manifold Approximation and Projection): it works in a similar way to t-SNE, but they differ in the way they build the graph. Clusters are more visually compact with more space between them. Faster than t-SNE**.**



Lun ATL, McCarthy DJ, Marioni JC (2016). "A step-by-step workflow for low-level analysis of single-cell RNA-seq data with Bioconductor." *F1000Res.*, **5**, 2122. doi: 10.12688/f1000research.9501.2.

Dimensionality reduction - visualization

Be very careful about distance in reduced space as they will be distorted

Minimum distance (min_dist) and **number of neighbors (n_neighbors)** are the most important parameters (hyperparameters)

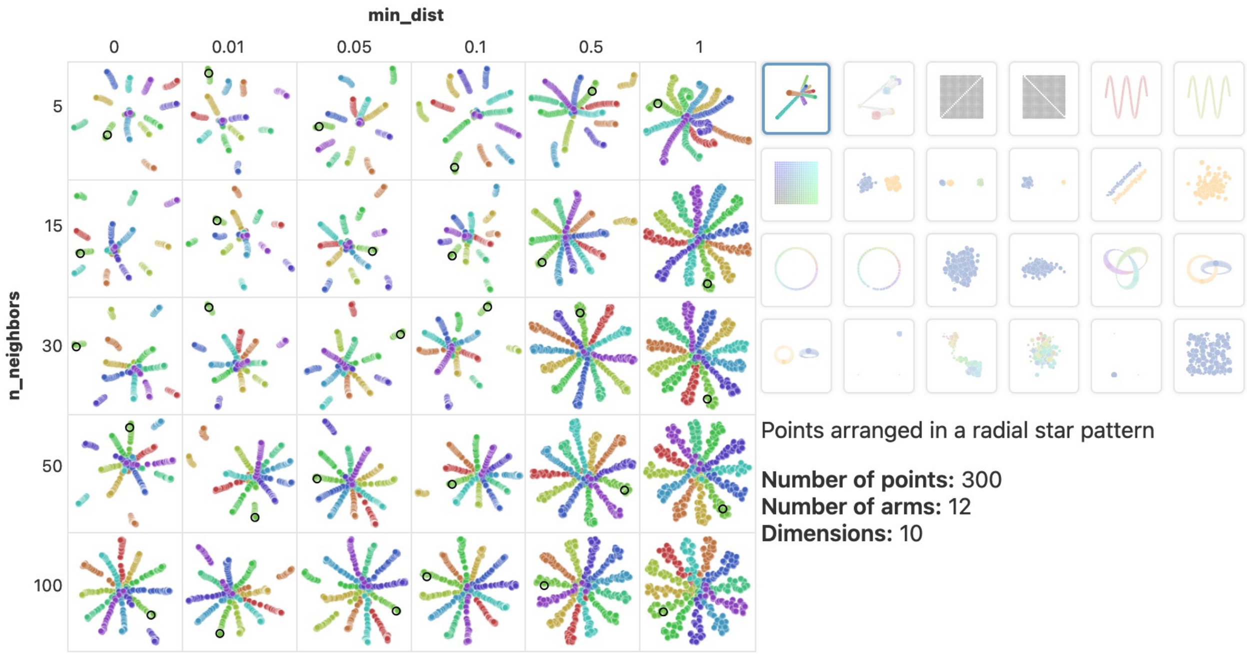


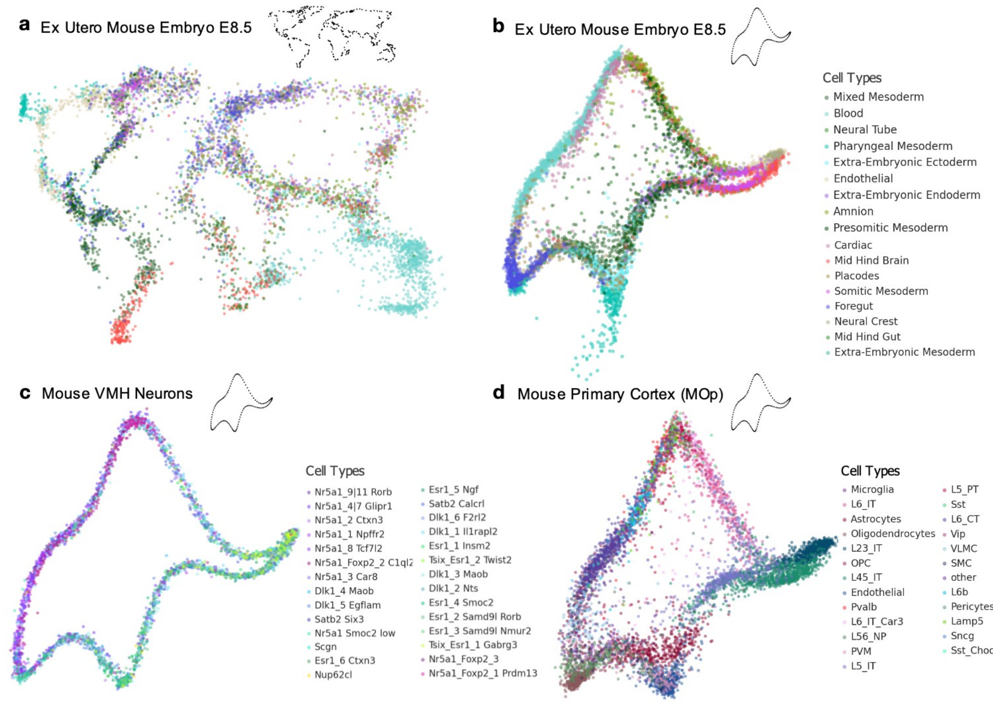
Figure 4: *UMAP projection of various toy datasets with a variety of common values for the* `n_neighbors` *and* `min_dist` *parameters.*

Further reading and parameter testing for UMAP: https://pair-code.github.io/understanding-umap/

Dimensionality reduction - visualization

UMAP/t-SNE may introduce severe distortions into high-dimensional data

- t-SNE and UMAP embeddings are not "canonical" representations but rather arbitrary projections that may mislead biological interpretation.
- The embeddings can create false impressions of cell mixing or separation that don't reflect the true high-dimensional relationships
- Nonetheless, UMAPs/t-SNE remain the de facto low-dimensionality projection used in single-cell data (for now)



Further reading: https://x.com/lpachter/status/1431325969411821572

Dimensionality reduction - visualization

Recommendations

- Select between 1,000 and 5,000 highly variable genes depending on dataset complexity.
- Dimensionality reduction methods should be considered separately for summarization and visualization
- PCA for general purpose summarization
- Use UMAP for visualization
- Use the UMAP plot for a diagnostic inspection of the cell clustering output, e.g., to check which clusters are close neighbors or whether a cluster can be split into further subclusters
- Be very **careful** about overinterpretations and 'tea-leaf reading' in UMAP or any other reduced space

Quiz

- Q1: Why do we need both PCA and UMAP/t-SNE in the analysis pipeline?
- Q2: Why is it risky to interpret distances in UMAP reduced dimensional space?

Quiz

- Q1: Why do we need both PCA and UMAP/t-SNE in the analysis pipeline?
  - They have different goals:
  - PCA: Computational workhorse for noise reduction and input to algorithms (clustering, integration) – unclear for 2D plotting
  - UMAP/t-SNE: Human-interpretable visualization tool for pattern recognition and biological insight

- Q2: Why is it risky to interpret distances in UMAP reduced dimensional space?
  - Dimensionality reduction algorithms like UMAP prioritize local structure preservation over global distances, meaning cluster proximity in 2D plots may not reflect true biological relationships in high-dimensional gene expression space.

Single-cell RNA workflow

Steps

- Generation of the count matrix
- Empty droplets detection
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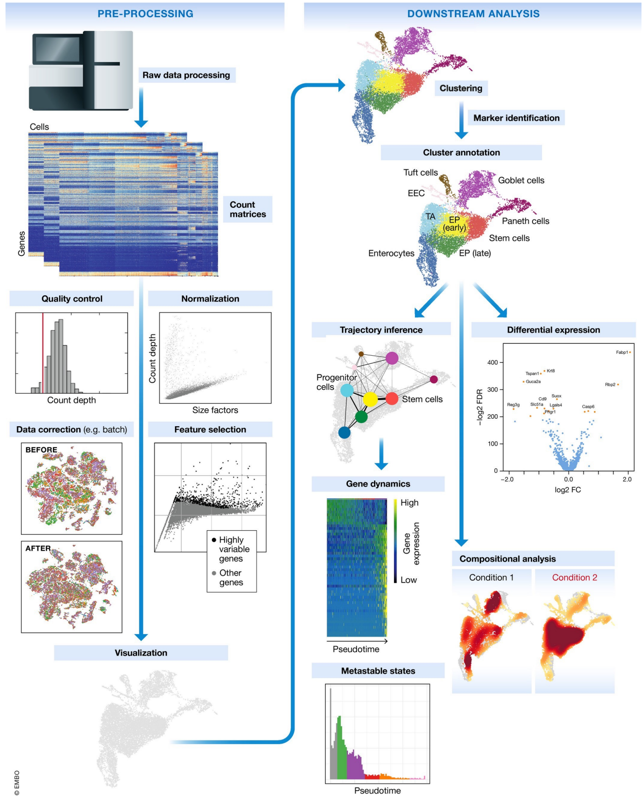
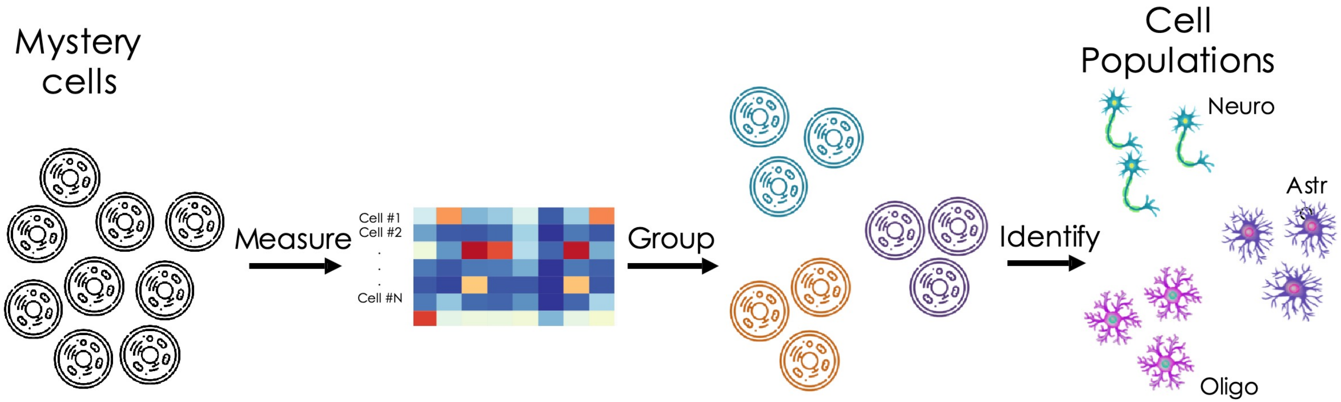
- Quality control: cells and transcripts
- Cell cycle
- Normalization
- Feature selection
- Dimensionality reduction
- Clustering



Cell Clustering – The mission

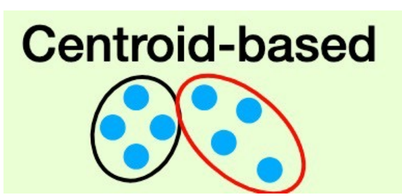


- Clusters allow us to infer the identity of member cells
- Clusters are obtained by grouping cells based on the similarity of their gene expression profiles
- Expression profile similarity is determined via distance metrics, which often take dimensionality-reduced representations as input

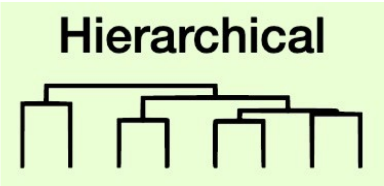
Clustering

Two approaches exist to generate cell clusters from these similarity scores:

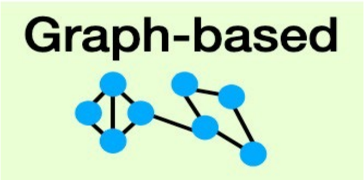
1. Clustering algorithms: Work on a distance matrix. Cells are assigned to clusters by minimizing intra cluster distances or finding dense regions in the reduced expression space.
2. Community detection: Build a graph where each node is a cell that is connected to its k nearest neighbours. Edges are weighted based on the similarity between the cells involved. Identify "communities" of cells that are more connected to cells in the same community than they are to cells of different communities.



kmeans

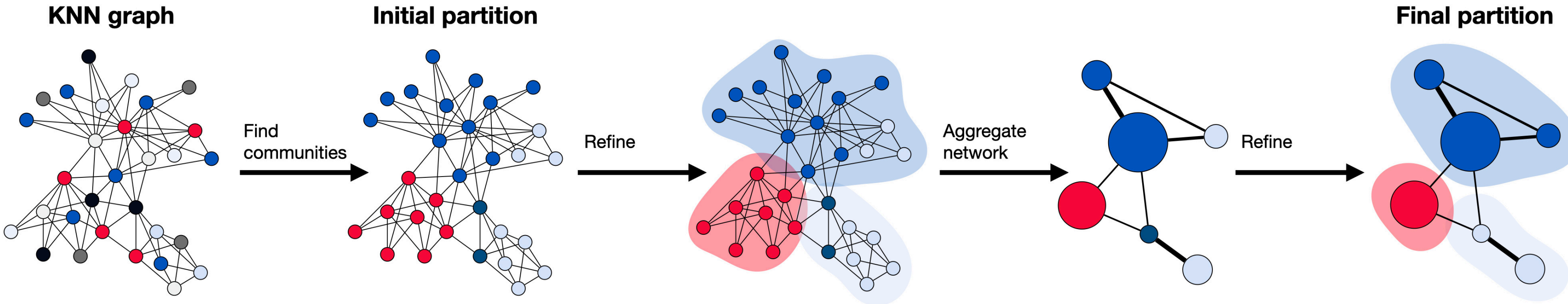


Ward



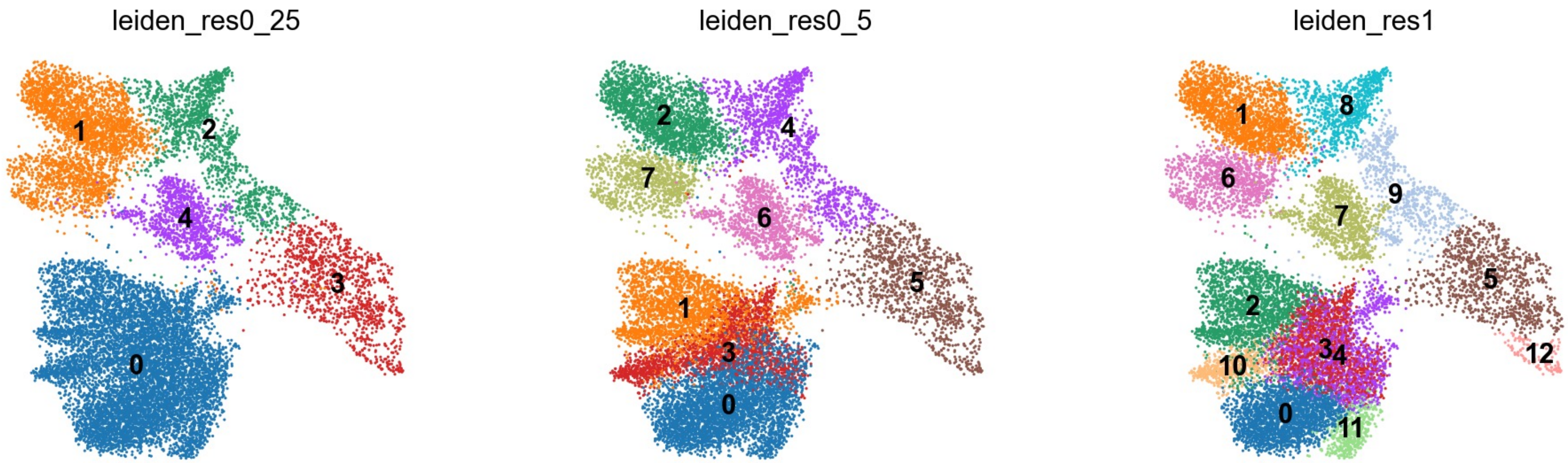
kNN

Graph-based clustering



The **Leiden** algorithm computes a clustering on a KNN graph obtained from the PC reduced expression space. It starts with an initial partition where each node from its own community. Next, the algorithm moves single nodes from one community to another to find a partition, which is then refined. Based on a refined partition an aggregate network is generated, which is again refined until no further improvements can be obtained, and the final partition is reached

Clustering – resolution parameter



Resolution parameter can be tuned to change the number of clusters

Clustering Evaluation

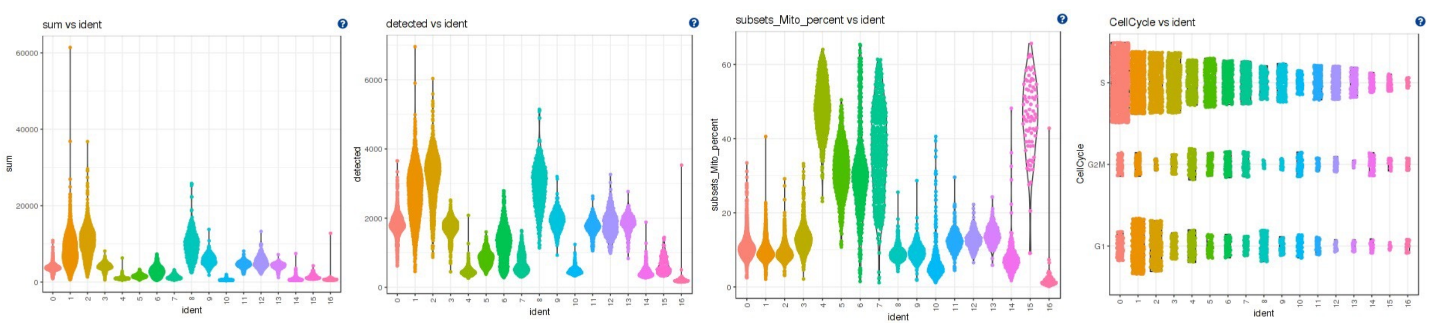
Graph-based (kNN) clustering outperforms the traditional clustering algorithms.

- It is faster → only requires a k-nearest neighbor search that can be done in log-linear time on average.

- avoids making strong assumptions about the shape of the clusters or the distribution of cells within each cluster.

- reduces the risk of generating many uninformative clusters consisting of one or two outlier cells.

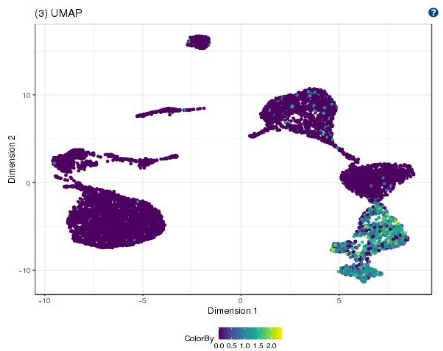
Assessing clusters: Segregation of clusters by various sources of uninteresting variation



- The metrics should be relatively even across the clusters.
- Segregation by cell cycle can be regressed (but I don't advise it)
- Keep an eye on clusters that exhibit a big difference on the other metrics and check if cell types explain the differences.

Clustering Assessment

Assessing clusters: Exploring known cell type markers

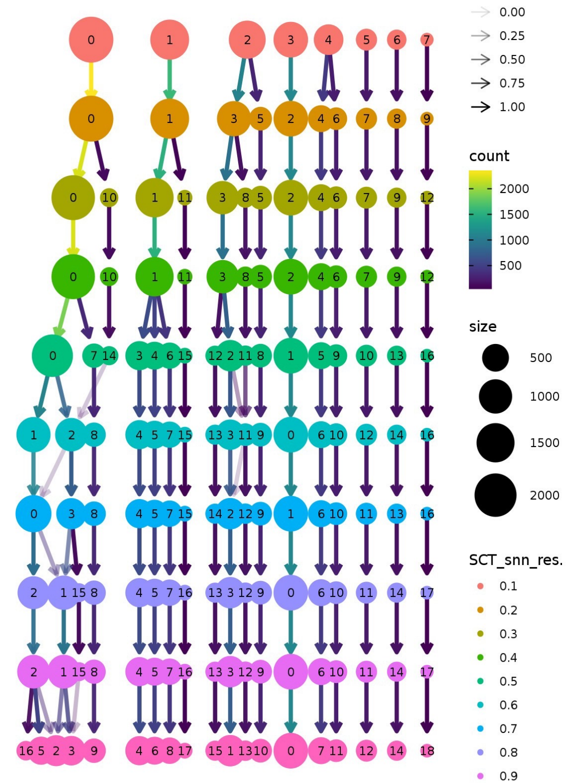


T-Cell marker CD3D

When clusters appear to contain two separate cell types:
- Increase clustering resolution.
- Increase number of PCs.
- Subclustering

Recommendations

- Have a good idea of your expectations for the cell types to be present and a handful of marker genes for these cell types
- Identify any junk clusters for removal



Quiz

- Q1: How do you choose the appropriate resolution for clustering?
- Q2: How to know when we have unstable / non-relevant clusters?

Quiz

- Q1: How do you choose the appropriate resolution for clustering?
  - Resolution should match the expected granularity of cell types, with higher resolution potentially splitting major cell types into subtypes. There isn't one optimal solution and will depends on your biological question.

- Q2: How to know when we have unstable / non-relevant clusters?
  - Warning signs include: extreme QC metric values, very few cells (<10), no distinguishing marker genes, high doublet scores, or disappearance when changing minor parameters.

Single-cell RNA workflow

Steps

- Generation of the count matrix
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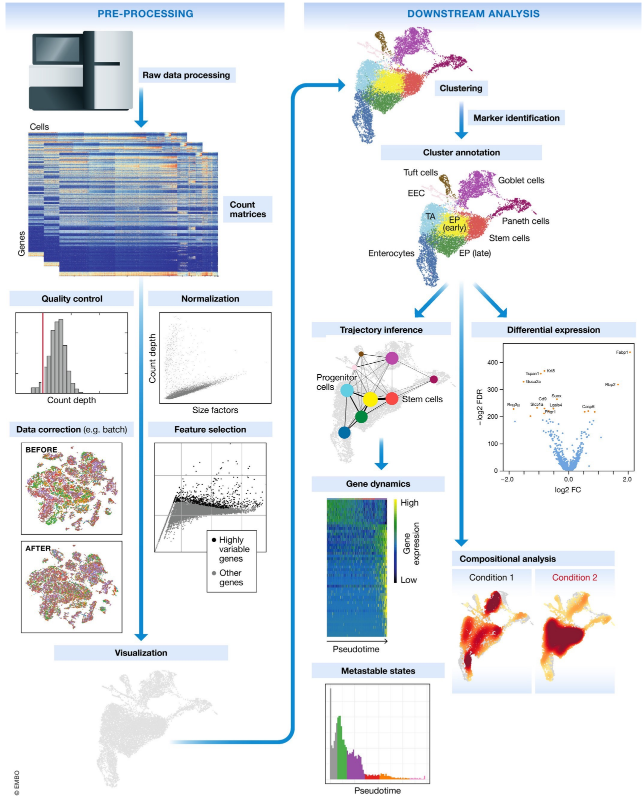
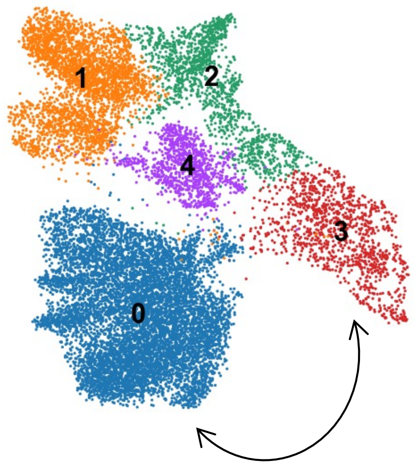
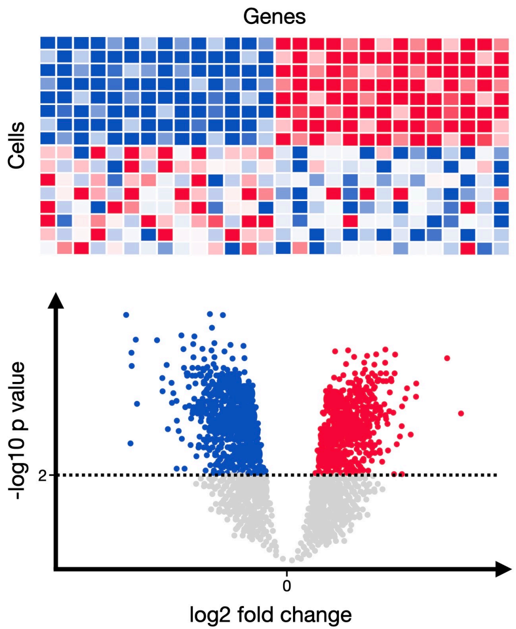
- Ambient RNA removal
- Quality control: cells and transcripts
- Cell cycle
- Normalization
- Feature selection
- Dimensionality reduction
- Clustering
- Cluster markers / differential expression



Marker Identification



Markers?



Markers identification

Motivation

- Identify the genes that drive separation between clusters.
- Assign biological meaning to each cluster based on their functional annotation.

Identification of marker genes is usually based around the retrospective detection of differential expression between clusters.

Several different statistical tests are available to quantify the differences in expression profiles, and different approaches can be used to consolidate test results into a single ranking of genes for each cluster.

Common types of marker identification

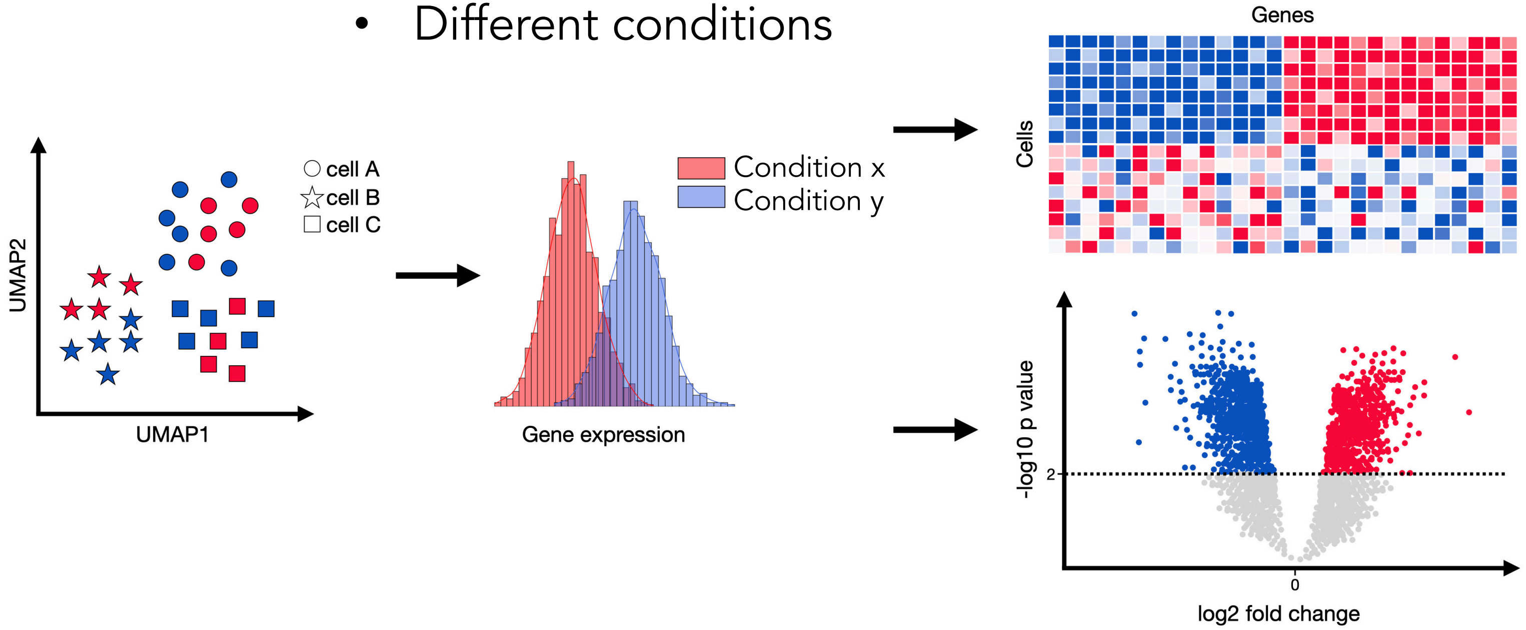
- Identification of all markers for each cluster: compares each cluster to the average of **all other cells**

- Marker identification between specific clusters: explores differentially expressed genes between **selected clusters**

gene	cluster	pct.1	pct.2	avg_logFC	p_val_adj
All	All	All	All	All	All
GNLY	0	0.989	0.372	2.18443834029039	0
NKG7	0	0.999	0.451	1.81947742147385	0
SPON2	0	0.905	0.091	1.5196707196473	0
GZMB	0	0.921	0.076	1.4135341482064	0
KLRD1	0	0.972	0.137	1.40943326896164	0
FGFBP2	0	0.86	0.069	1.3709066233598	0
PRF1	0	0.954	0.149	1.34123090161696	0
CST7	0	0.942	0.178	1.25796818689777	0
GZMA	0	0.927	0.161	1.15391041555303	0
CTSW	0	0.895	0.136	1.13870105926787	0
CD247	0	0.926	0.159	1.11380453964707	0

Differential Expression

We can also compare:
- Different clusters
- Different conditions



Markers identification

Three aspects should be noted when detecting marker genes:

- The **P-values** obtained for marker genes are based on the assumption that the obtained cell clusters represent the biological ground truth
- Clusters and marker genes are typically determined based on the same gene expression data. **Inflated** p-values occur.
- Marker genes differentiate a cluster from others in the dataset and are thus dependent not only on the cell cluster, but also on the dataset composition

Recommendations

- Do not use marker gene P-values to validate a cell-identity cluster, especially when the detected marker genes do not help to annotate the community. **p-values may be inflated**

- Note that marker genes for the same cell-identity cluster may differ between datasets purely due to dataset cell type and state compositions (=what cell population is being compared to what)

Quiz

- Q1: What is differential gene expression and in which cases are we interested in testing for it?
- Q2: What is the 'multiple testing' problem and how can it be eluded?

Quiz

- Q1: What is differential gene expression and in which cases are we interested in testing for it?
  - Differential gene expression (DEG) analysis identifies genes that show statistically significant differences in expression levels between biological conditions / states (like treated vs control, different cell types, or disease states).

- Q2: What is the 'multiple testing' problem and how can it be eluded?
  - Testing thousands of genes simultaneously inflates false positive rates; controlled using adjustment methods like Benjamini-Hochberg (FDR) correction or Bonferroni correction to maintain overall error rates.

Single-cell RNA workflow

Steps

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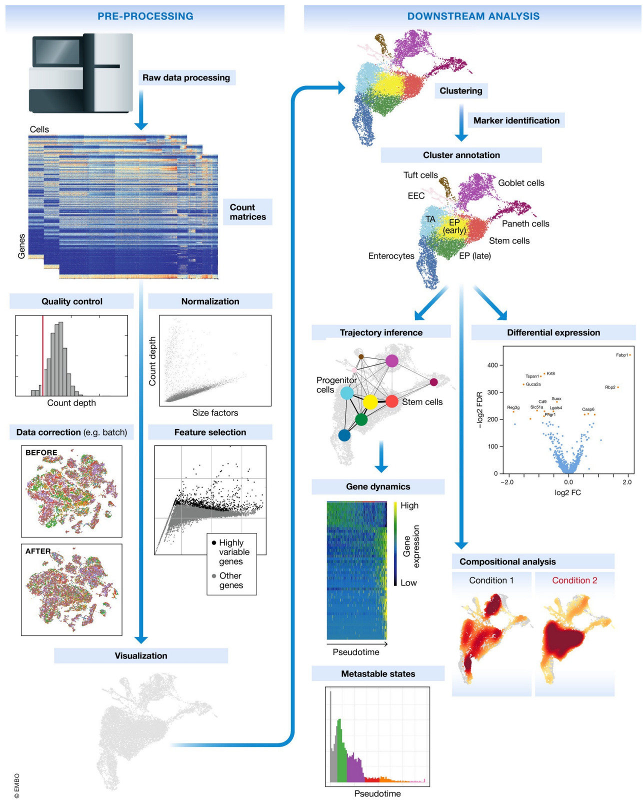
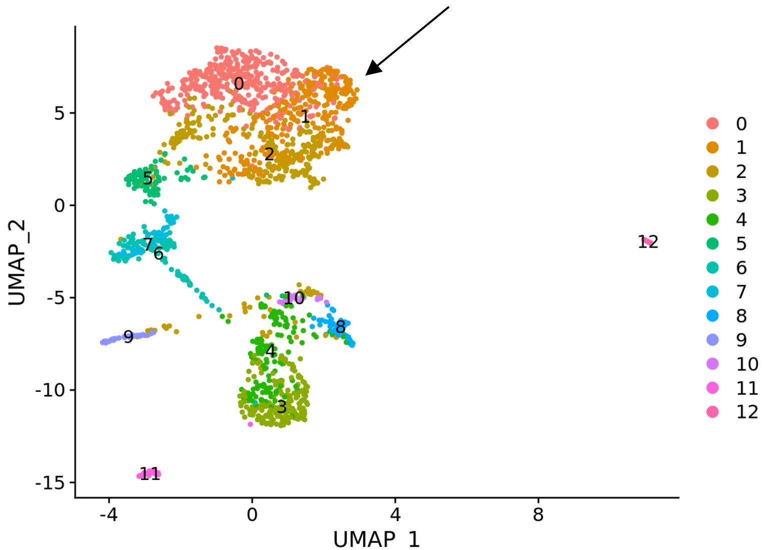
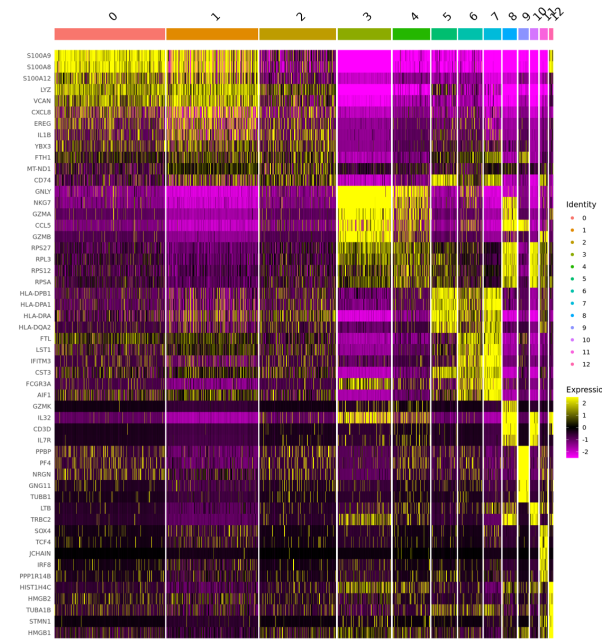
- Feature selection
- Dimensionality reduction
- Clustering
- Cluster markers / differential expression
- Cell type annotation



Cell type is the major determinant of gene expression

Which cell types are these cells?





Underlying assumption: cell cluster ~ cell type

Cell type annotation

Motivation: facilitate and speed up the interpretation of scRNA-seq data

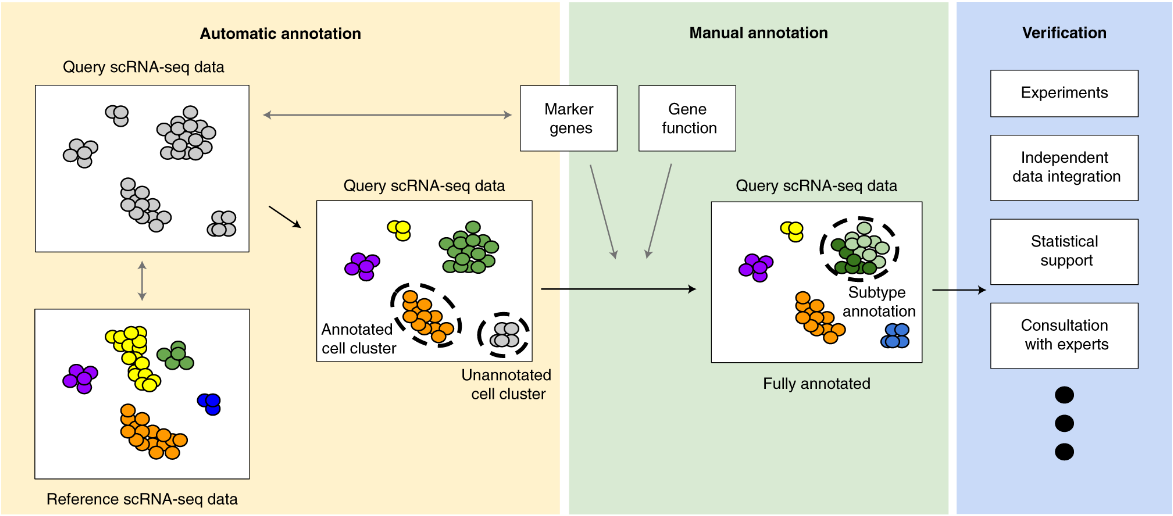
Several approaches

I. per cell:

a) **Reference-based:** Assigning labels from reference data: compare the single-cell expression profiles with previously annotated reference datasets **(preferred method if a good reference is available)**

b) **Marker-based:** Assigning labels from gene sets: identify sets of marker genes that are highly expressed in each individual cell

II. per cluster: Assigning labels from markers: perform a gene set enrichment analysis on the marker genes defining each cluster. E.g. use gene sets from the Gene Ontology (GO) or the Kyoto Encyclopedia of Genes and Genomes (KEGG) collections



Cell type annotation tools

Cell markers databases

- Cellmarker: link
- Cellguide from cell x gene: link
- scType: https://sctype.app/database.php
- PanglaoDB: link
- **Celltypist**
- CellKB

References / atlases

- cellxgene-census 120M cells
- Human Cell Atlas mapping (healthy) with CuratedAtlasQueryR
- Reference based atlases from Carmona lab - cancer + virus
- DISCO for immune / cancer
- Cancer Atlas - Weismann
- TISCH - Cancer specific
- Broad institute 42M cells
- EBI 10M cells
- Celltypist organs - Wellcome trust
- SingleCellAtlas - Stockholm
- Archmap
- Allen immune atlas

Cell type annotation

Recommendations

- For large datasets that contain many clusters, **automated** cell-identity annotation can be used to coarsely label cells and identify where sub-clustering may be needed. Marker genes should be calculated for the dataset clusters and compared to known marker gene sets from the reference dataset or literature

- Confirm the automatic annotation with **multiple** relevant single-cell references if possible

- For smaller datasets and datasets that lack reference atlases, **manual** annotation will suffice

Quiz

- Q1: How can you validate cell type annotations?
- Q2: What are the limitations of automated annotation tools?

Quiz

- Q1: How can you validate cell type annotations?
  - Cross-reference with multiple marker databases, examine expression of canonical markers, compare with orthogonal experimental data (FACS, immunostaining), and verify biological pathway coherence.

- Q2: What are the limitations of automated annotation tools?
  - May miss novel cell states, struggle with transitional populations, depend heavily on reference quality, and can misclassify cells in disease or perturbed conditions not well-represented in training data.

Hands-On analysis with Seurat in Renku

Dimensionality Reduction

Single-cell RNA workflow

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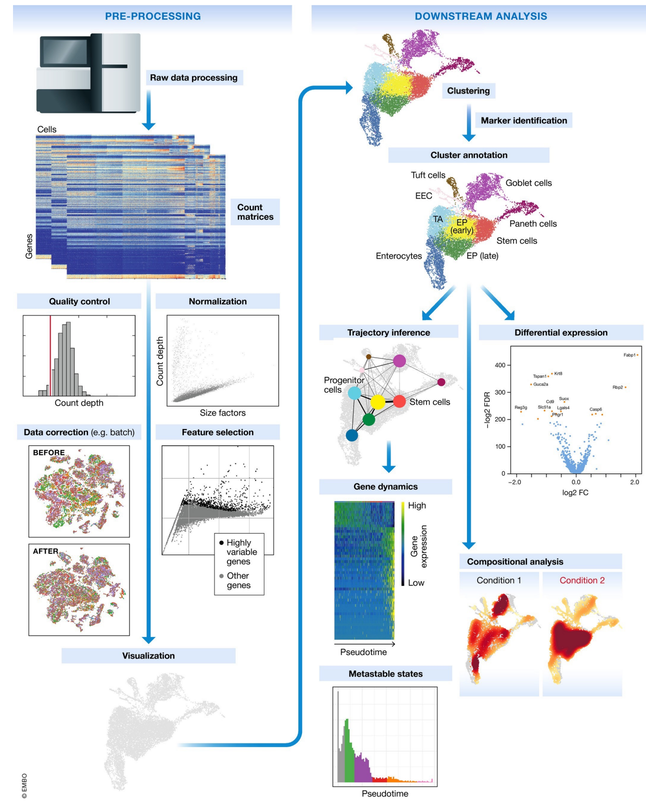
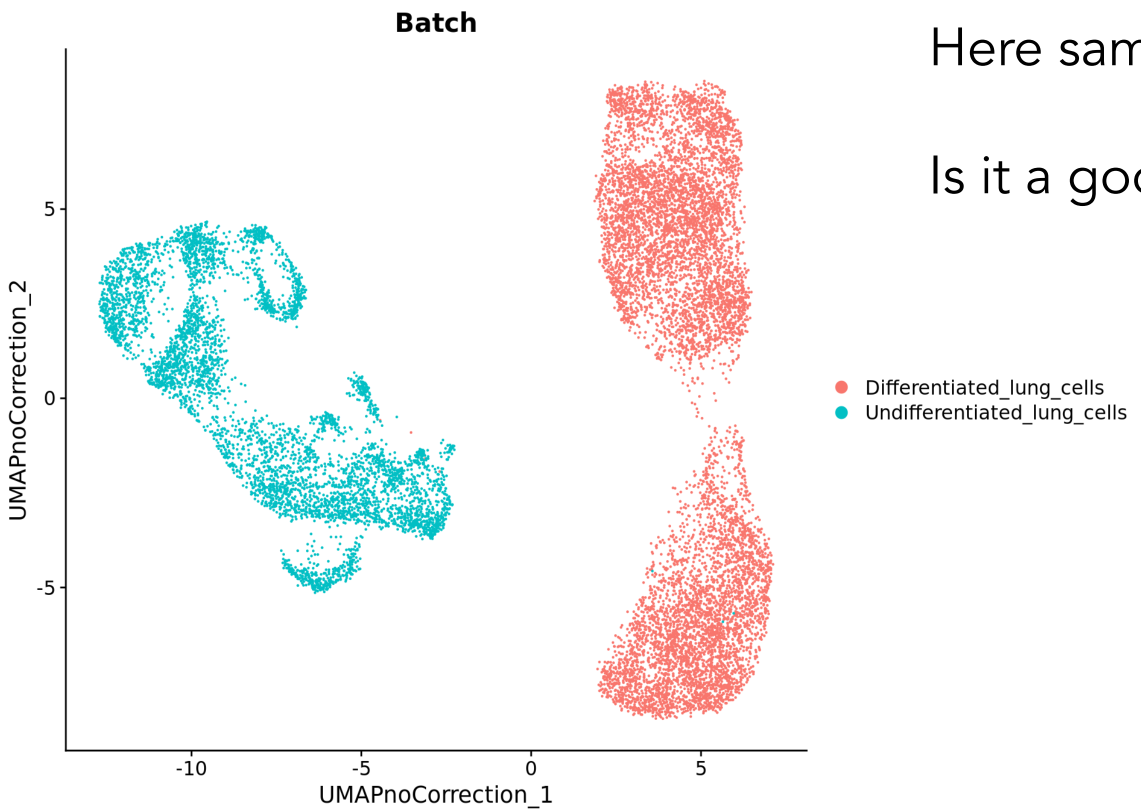
- Dimensionality reduction
- Clustering
- Cluster markers
- Cell type annotation
- Integrating and comparing samples



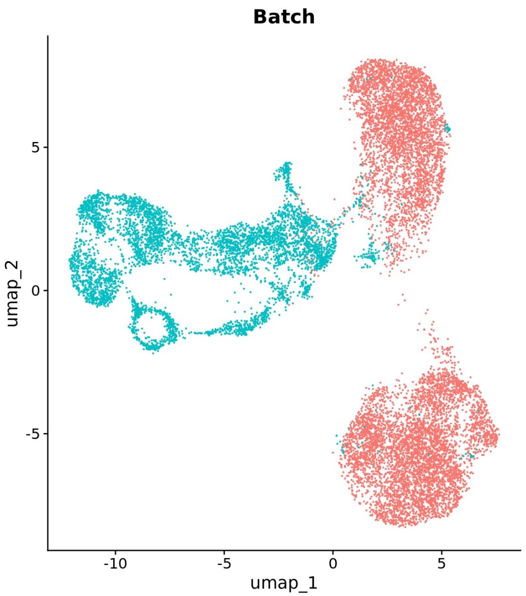
- No Correction



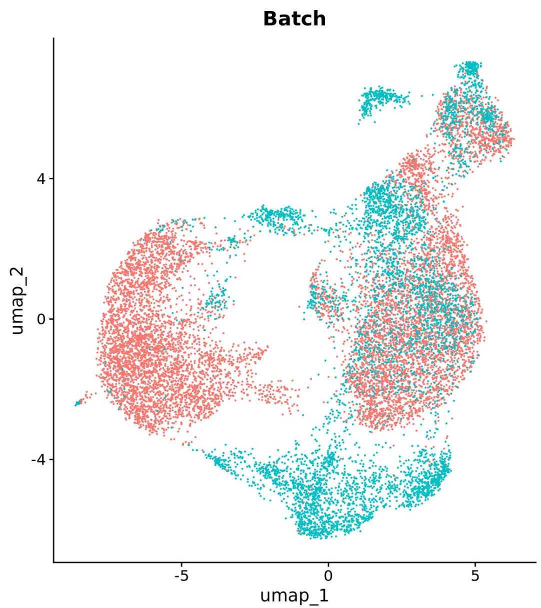
Here samples are split by batch

Is it a good sign or not?

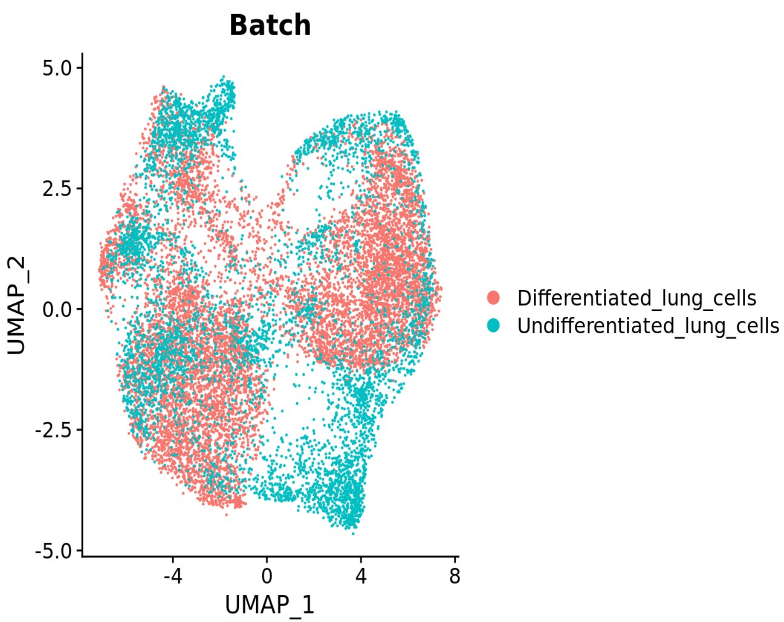
- Harmony



- RPCA

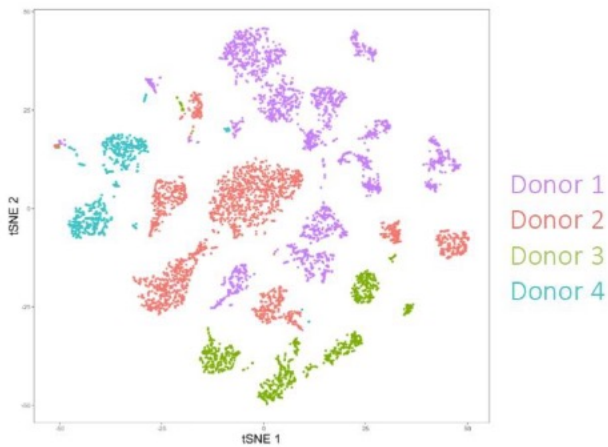


- CCA

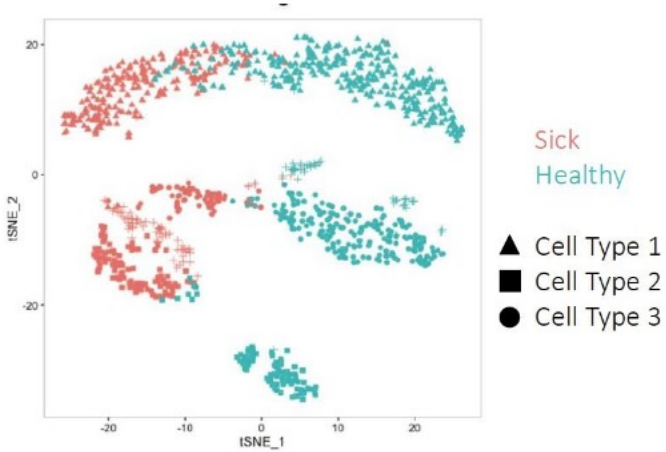


Integrating samples

Why do we want to integrate?



Same tissue from different donors



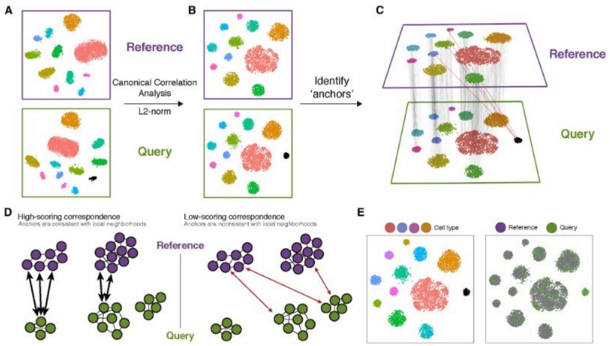
Cross condition comparisons

Integrating samples

What are batch effects?
Batch effects happen when cells are processed in separate groups, for example on different chips, in different sequencing lanes, or at different times. These differences can introduce unwanted variability in the data.

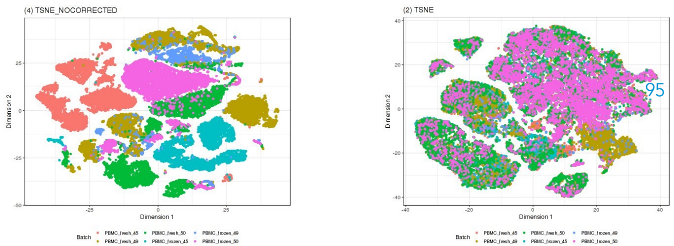
Correcting batch effects in the same experiment
When samples come from the same experiment, **linear methods** can be used to correct batch effects.
These methods assume that you either know the composition of the cell populations or that it remains the same across different batches.
This approach is often used in bulk RNA-seq, where overall cell population composition is relatively stable.

Correcting batch effects across different experiments or labs
When datasets come from different experiments or different labs, **nonlinear methods** are used.
These methods do not require prior knowledge about the cell population makeup and can handle more complex differences.



Comprehensive integration of single cell data
Tim Stuart, Andrew Butler, Paul Hoffman, Christoph Hafemeister, Efthymia Papalexi, William M. Mauck III, Marlon Stoeckius, Peter Smibert, Rahul Satija

Diagnosing batch effects



Integration analysis: Confounders and batch effects

Technical Variability

•Variations in sample quality or processing

•Differences in library preparation or sequencing technology

These technical “batch effects” can interfere with downstream analyses.

Biological Variability

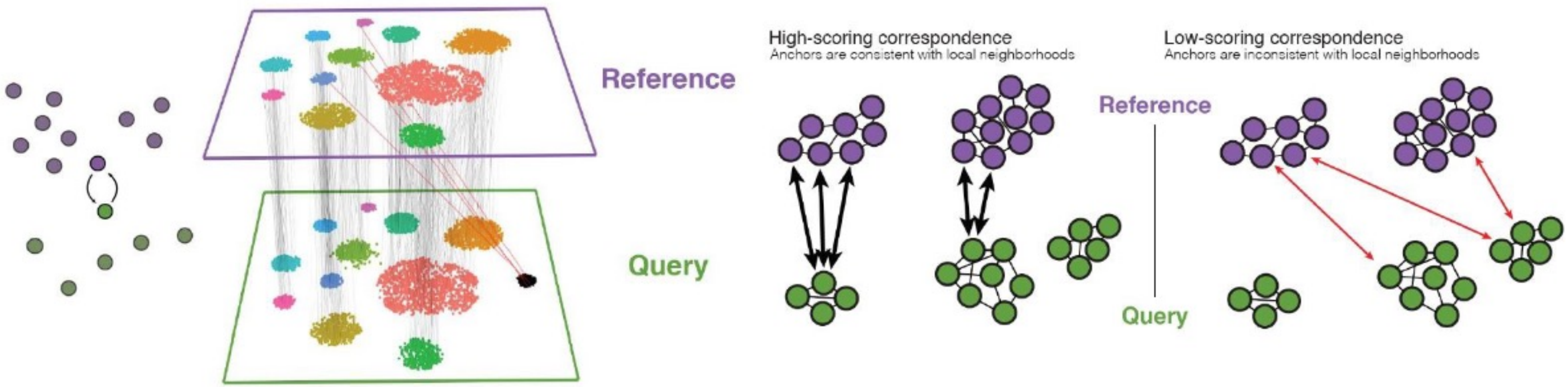
•Differences among patients

•Evolutionary differences (e.g., cross-species analyses)

These biological “batch effects” complicate comparisons of scRNA-seq data.

Integration analysis tools: Basic idea

1. Find corresponding cells across datasets (= 'anchors' by computing a distance between cells in a certain space)
2. Compute a data adjustment based on correspondences between cells
3. Apply the adjustment



Integrating samples

Global models

- Fit regression model with batch effect covariate

Residuals (often using linear regression):

$$\hat{n}_{gc} = f_D(B_c, \dots)$$

$$r_{gc} = n_{gc} - \hat{n}_{gc} = n_{gc} - (\beta_0 + \beta_1 B_c)$$

in linear model case

Example:
sc.tl.regress_out()

Correct for fitted batch effect:

$$n_{gcb} = \alpha_g + X\beta_g + \gamma_{gb} + \delta_{gb}\epsilon_{gcb}$$

bio design matrix

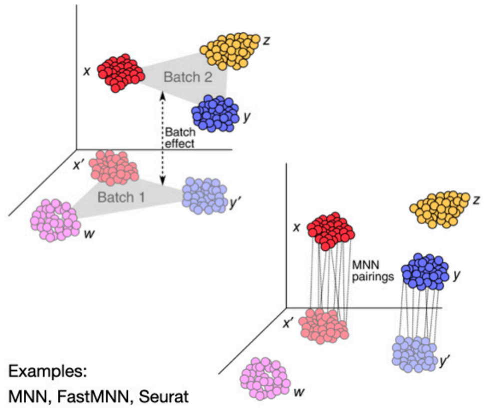
additive batch effect

multiplicative batch effect

Example:
ComBat - ***scanpy.pp.combat()***

Linear embedding models

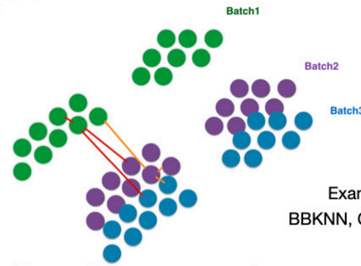
- Project cells into low dimensional embedding
- find most similar cells in other batch e.g., using mutual nearest neighbours (MNNs)
- Use MNNs as anchors to calculate a correction vector



Examples:
MNN, FastMNN, Seurat v3, Scanorama

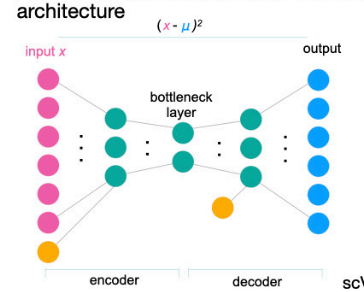
Graph-based methods & Deep learning

Enforce graph connections between different batches



Examples:
BBKNN, Conos

Add condition node into auto-encoder architecture

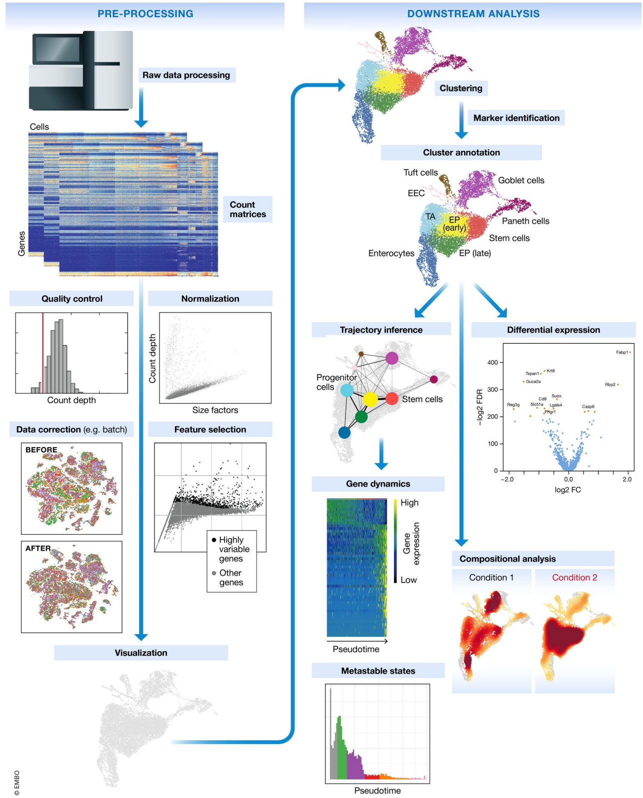


Examples:
scVI, trVAE, SAUCIE

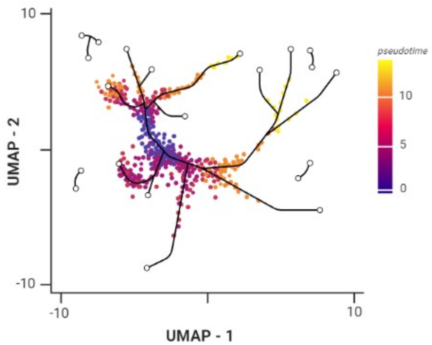
Single-cell RNA workflow

Steps

- Generation of the count matrix
- Empty droplets detection
- Doublet detection
- Ambient RNA removal
- Quality control: cells and transcripts
- Cell cycle
- Normalization
- Feature selection
- Dimensionality reduction
- Clustering
- Cluster markers
- Cell type annotation
- Integrating and comparing samples



Possible analyses after cell annotation and DGE (Differential Gene Expression)



- Trajectory Analysis
- Cell-cell communication
- Metacell analysis
- Pathway analysis / Gene Regulatory Networks
- Velocity

Additional learning resources

- Seurat tutorial – pbmc 3k (good starting point)
- scRNA-seq tutorial on youtube – chatomics
- Orchestrating Single-Cell Analysis with Bioconductor
- Single-cell best practices (python-based)

Hands-On analysis with Seurat in Renku

Integration & Annotation

Supplementary Slides

Trajectory analysis

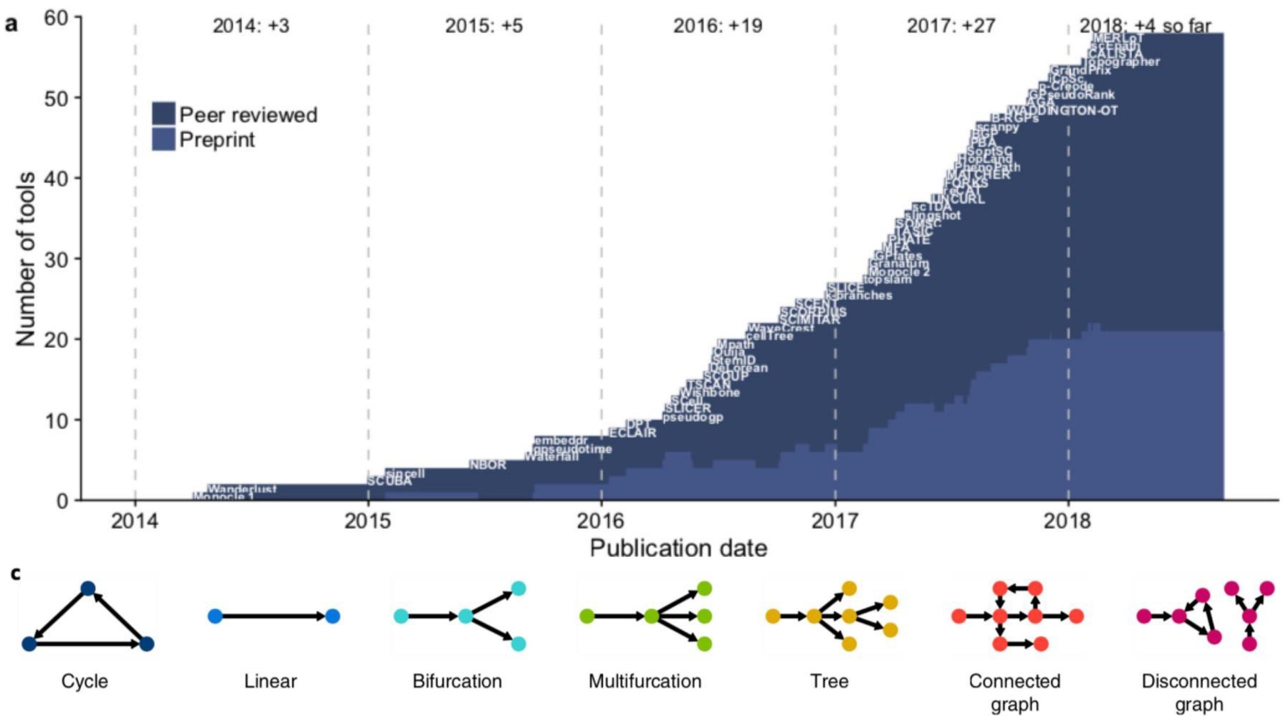
- **Why differences occur:** Cell-to-cell gene expression variations can be caused by dynamic processes, such as
    - Cell cycle changes
    - Cell differentiation
    - Responses to external stimuli
- **What trajectory inference does:**
    - Orders cells along a path or “pseudotime” axis to reveal how they transition between states.
    - Some methods project cells onto a single pseudotime axis, while others allow branching paths.
- **Why it’s useful:**
    - Helps identify the gene expression programs driving specific cell phenotypes.
    - Enables discussion on how conditions influence whether cells become more or less differentiated (e.g., comparing the number of cells at the start vs. end of the pseudotime axis).

Should you run trajectory inference?

- Are you sure that you have a developmental trajectory?
- Do you have intermediate states?
- Do you believe that you have branching in your trajectory?
- Do you have a time series experiment?
- Do you have a starting state or an ending state?

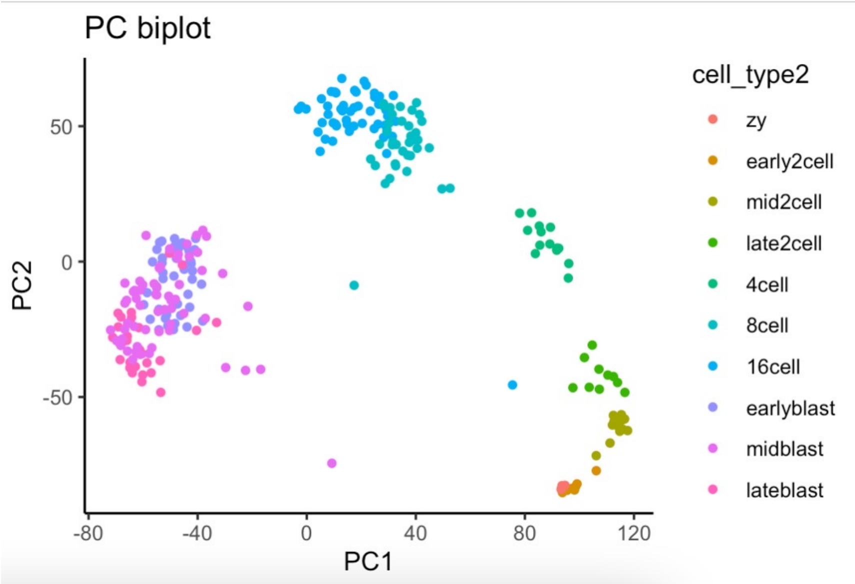
Be aware, any dataset can be forced into a trajectory without any biological meaning!

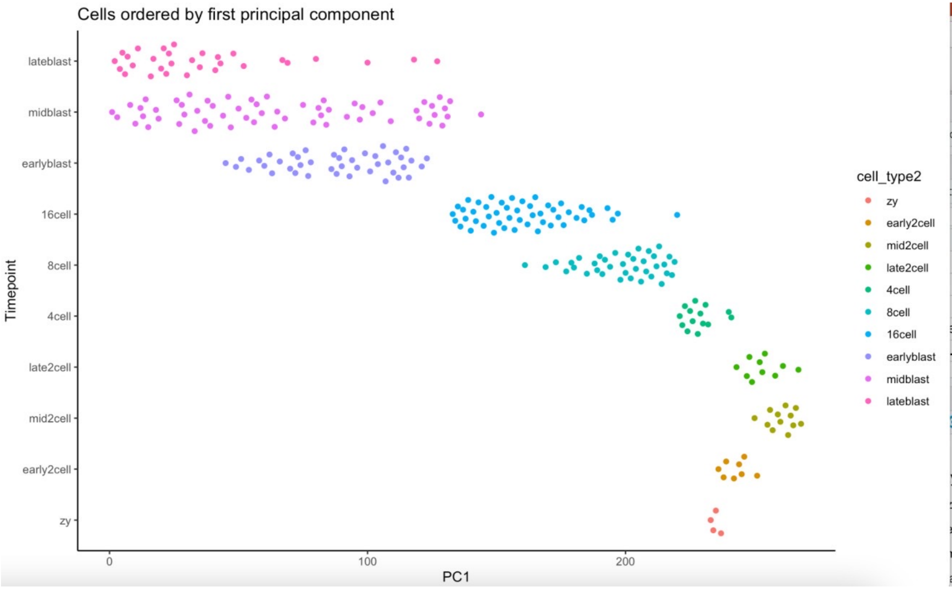
Different types of trajectory analysis



Saelens et al (2019) Nat Biotechnology

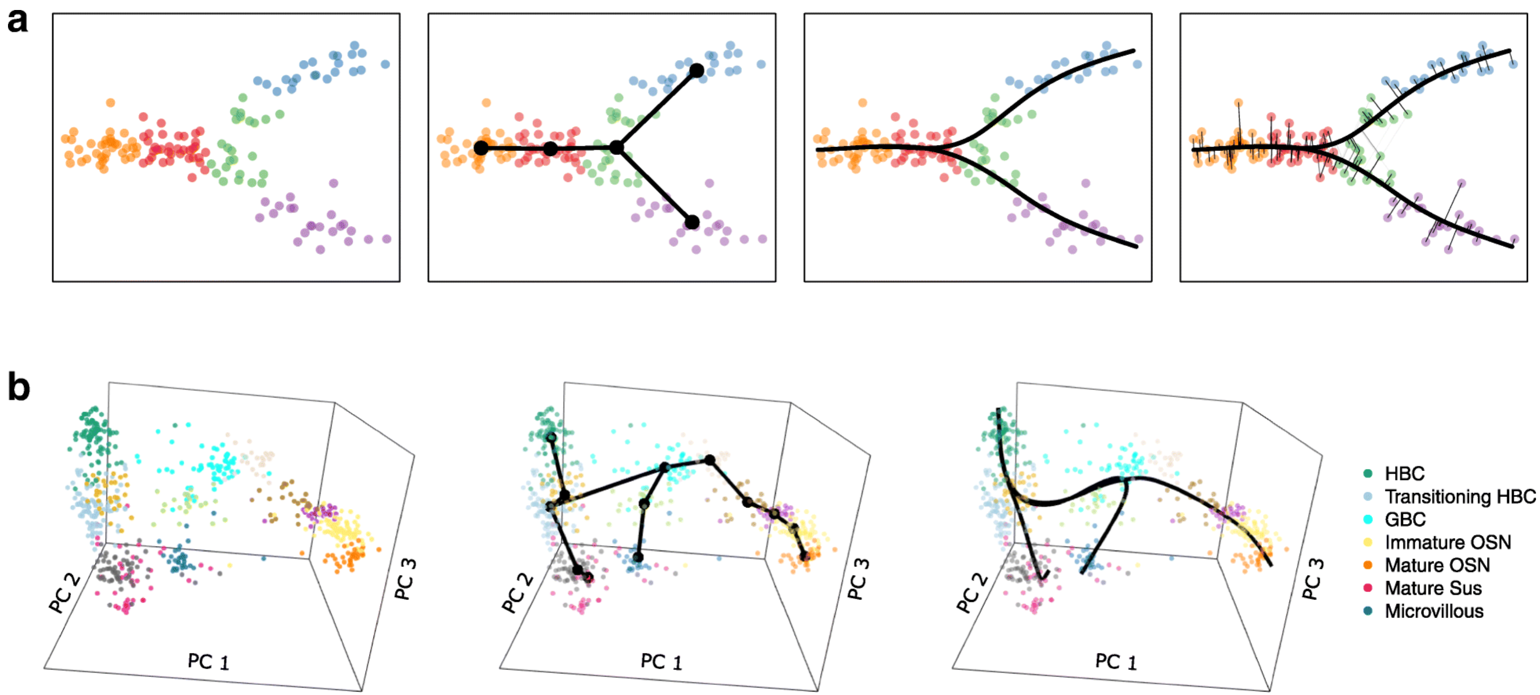
A trajectory could be to order the cells according to PC1



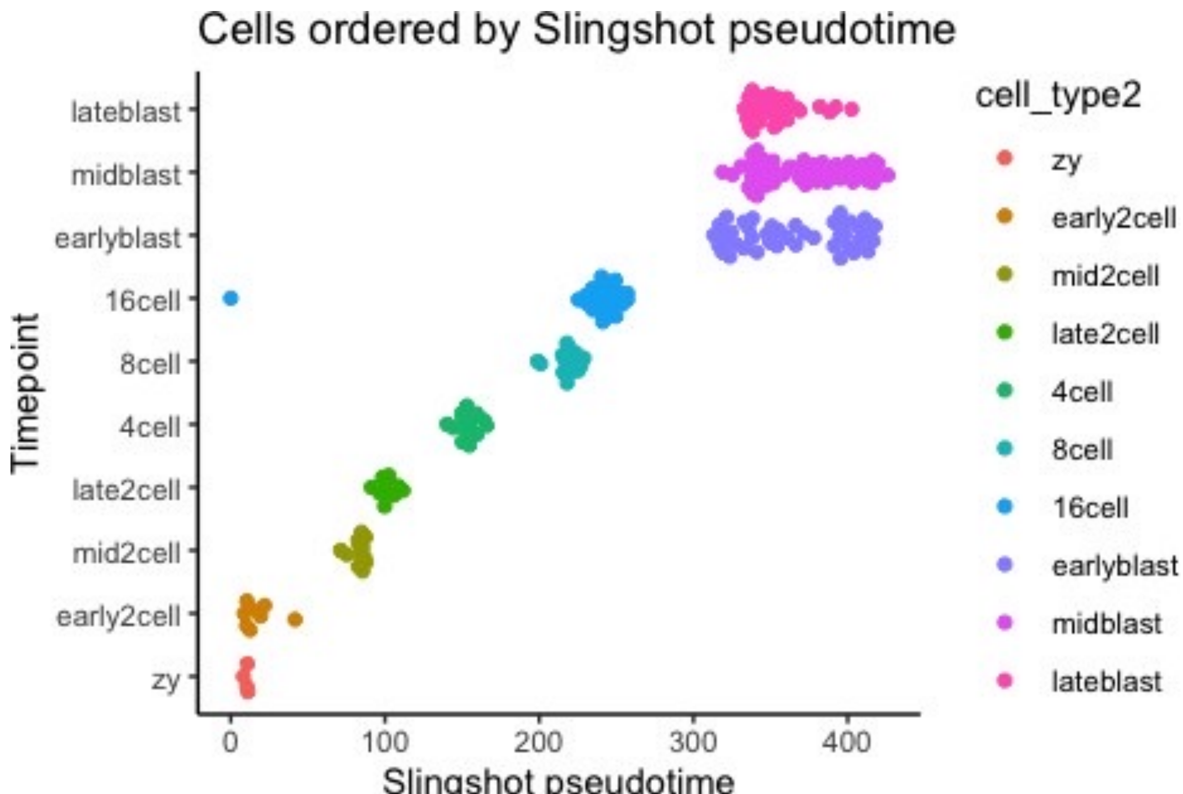


But can we do better ?

Slingshot (Street et al 2018)

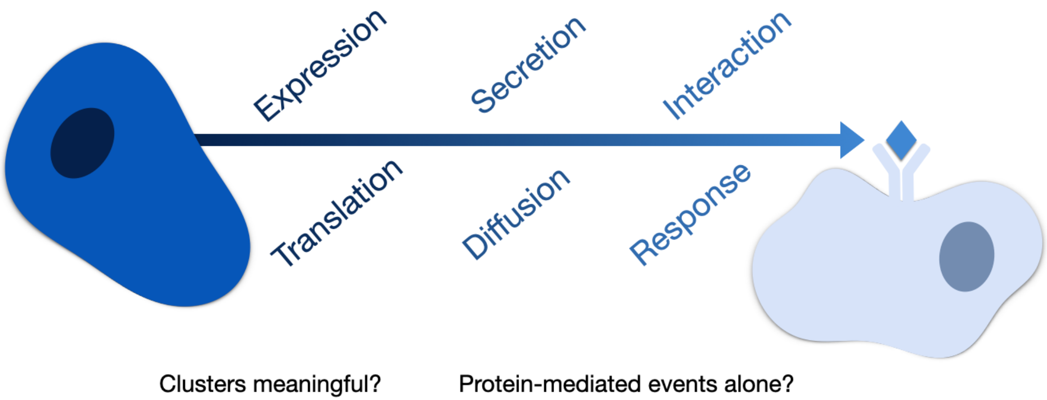


Slingshot on Deng data



Cell-cell Communication (CCC)

Which cells are 'talking' to each other?



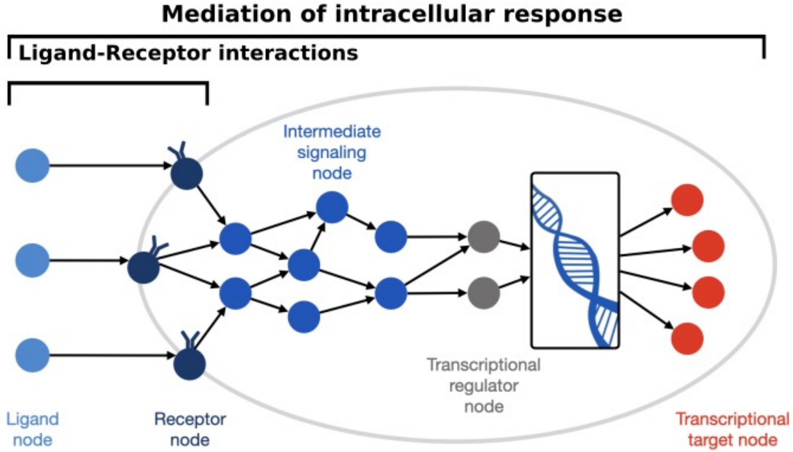
Cell-cell communication assumptions

- **CellphoneDB** - https://www.cellphonedb.org/ - online « clickable » Mirjana Efremova, Nat protocols, 2020.
- **NicheNet** – needs apriori knowledge, Robin Browaeys, Nat met, 2020.
- **CellChat**- http://www.cellchat.org/

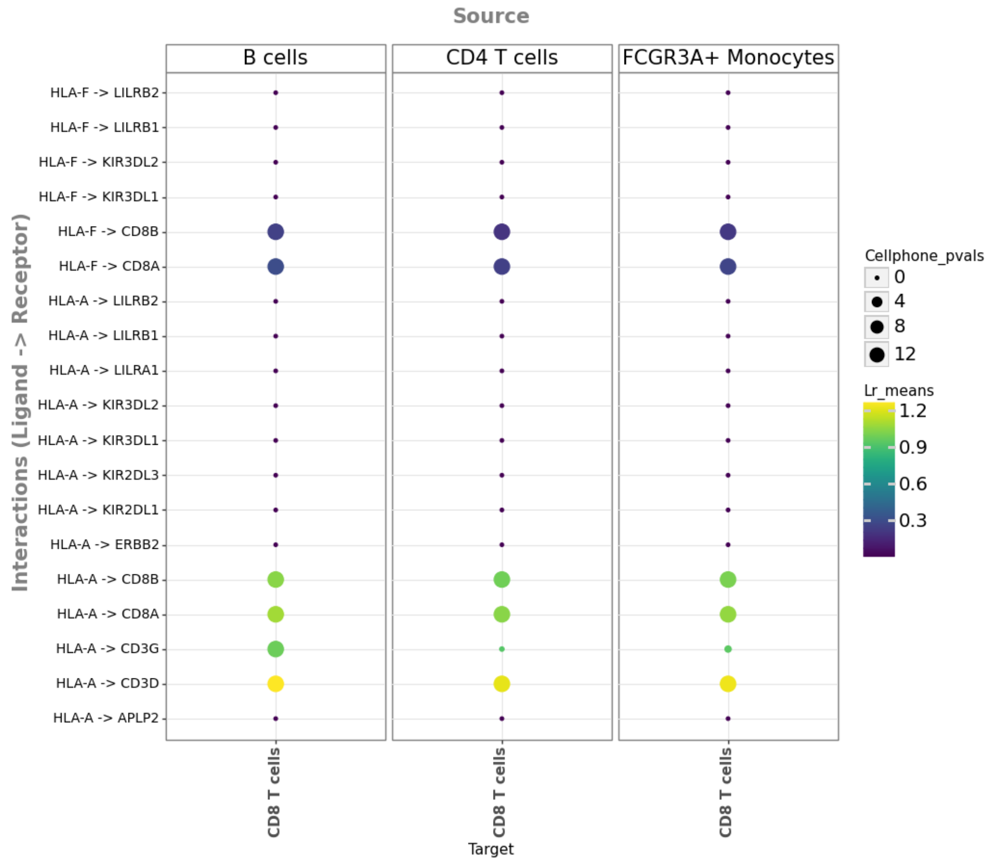
NicheNet- Ligand receptor analysis

- Question: In your analysis you have a certain list of differentially expressed genes. Can one associate a pair of Ligand and Receptor responsible for the change in expression observed?

- This is extremely useful as it will point biologist to possible pathways to target to block the changes observed and which cell types communicate.



NicheNet- Ligand receptor analysis



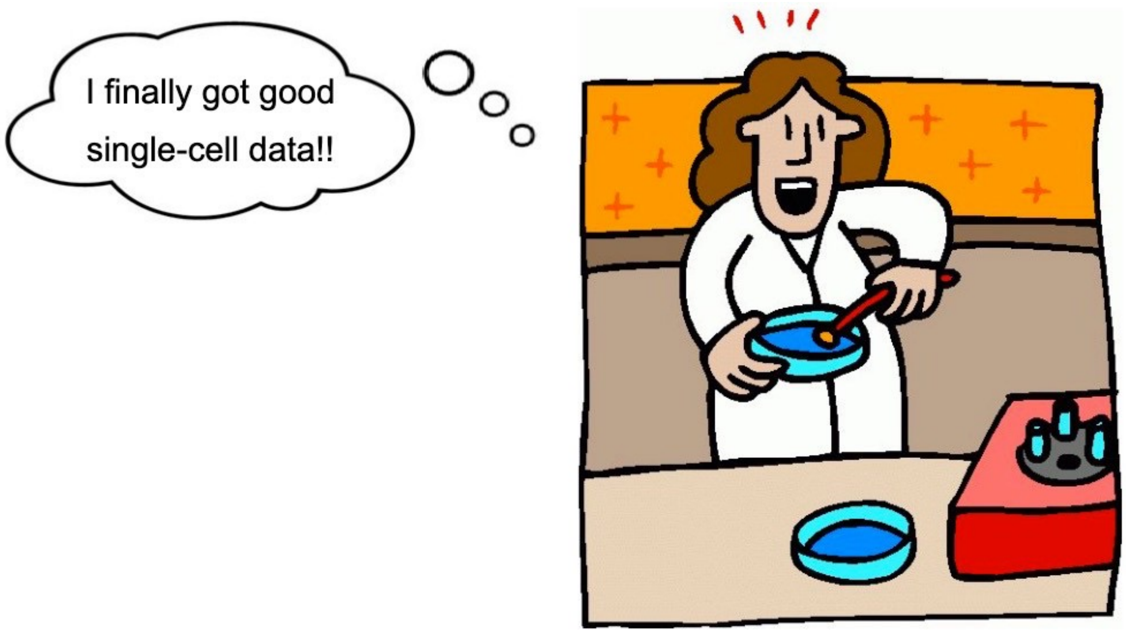
Quiz

- Q1:What are the limitations that come to mind when inferring CCC from single-cell transcriptomics data?

Quiz

- Q1:What are the limitations that come to mind when inferring CCC from single-cell transcriptomics data?
  - Spatial context loss: Dissociation removes information about which cells were actually neighbors
  - mRNA vs protein levels: Transcript abundance doesn't guarantee functional protein expression
  - Temporal dynamics missing: Single timepoint snapshots miss the dynamic nature of signaling

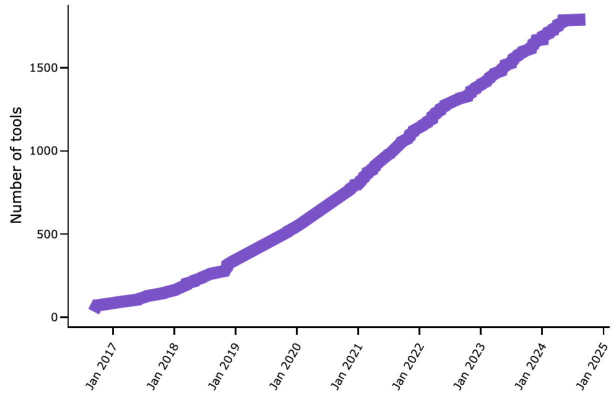
Tools for the analysis of single-cell RNA-Seq data



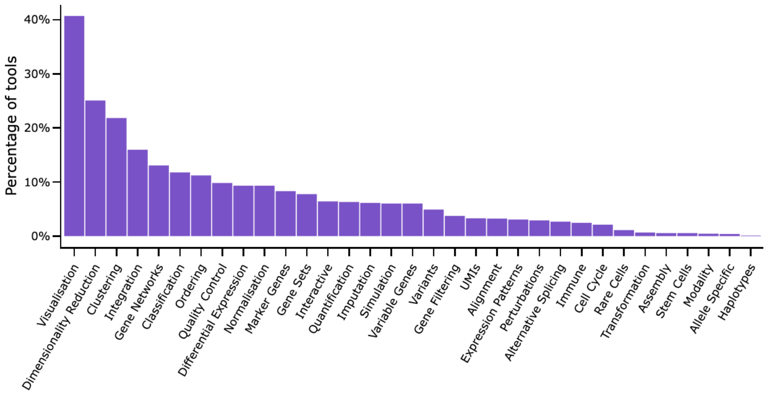
scRNA-tools Overview

https://www.scrna-tools.org/

The scRNA-tools database records details of software tools designed for analysing scRNA-seq data. Each tool is categorised according to the analysis tasks it can be used for.



We currently track 1791 tools...



...in over 30 categories

Tools to analyze Single-cell RNA-Seq data

- Open source software in R or Python: Provide a basis for the construction of analysis pipelines. Needs programming skills.

  - Seurat (Butler et al, 2018) => most used **R** package
  - scran/Bioconductor (Lun ATL et al, 2016) => **R**, similar to Seurat but less documented
  - Scanpy (Wolf et al, 2018) => **Python**

- Graphical user interface platforms: Users are often guided through prescribed workflows that facilitate the analysis, but also limit user flexibility.

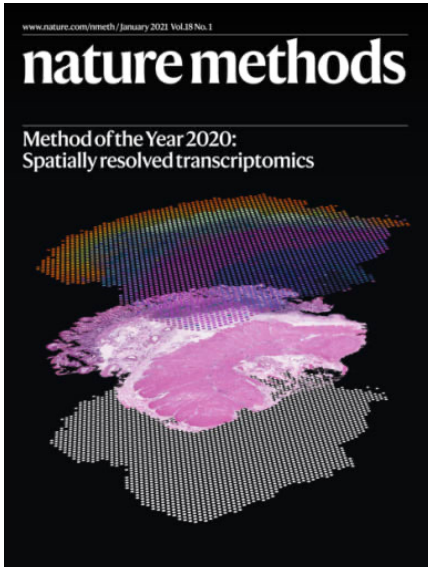
  - Loupe Browser fromr 10x Genomics
  - SUSHI (FGCZ)
  - [CellxGene]

Recent developments of the single-cell field

Some of the most recent development in the field include:

- Spatial transcriptomics
- Multi-modal datasets
- Transformer-based (AI) approaches:
  Geneformer, scGPT etc

  Goal: **Universal cell modelling,**
  cf paper [Bunne et al. 2024]



Additional Slides – Structure of a Seurat Object